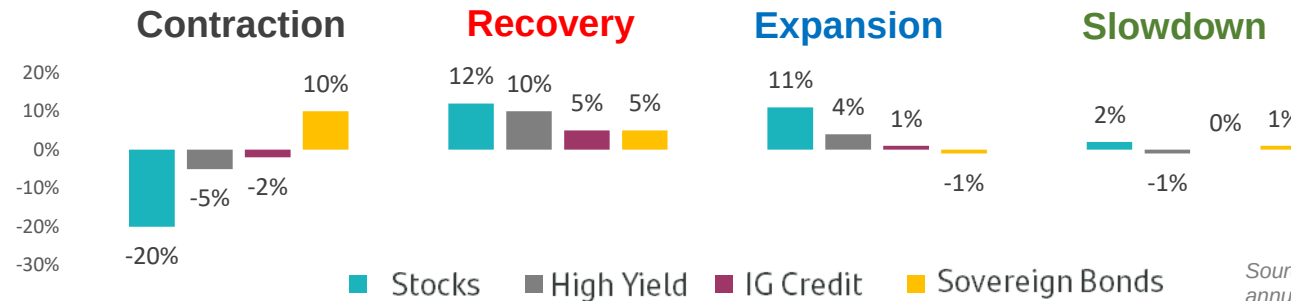




Economic Regime-Based Asset Allocation

“Put data, not emotions, behind your money”

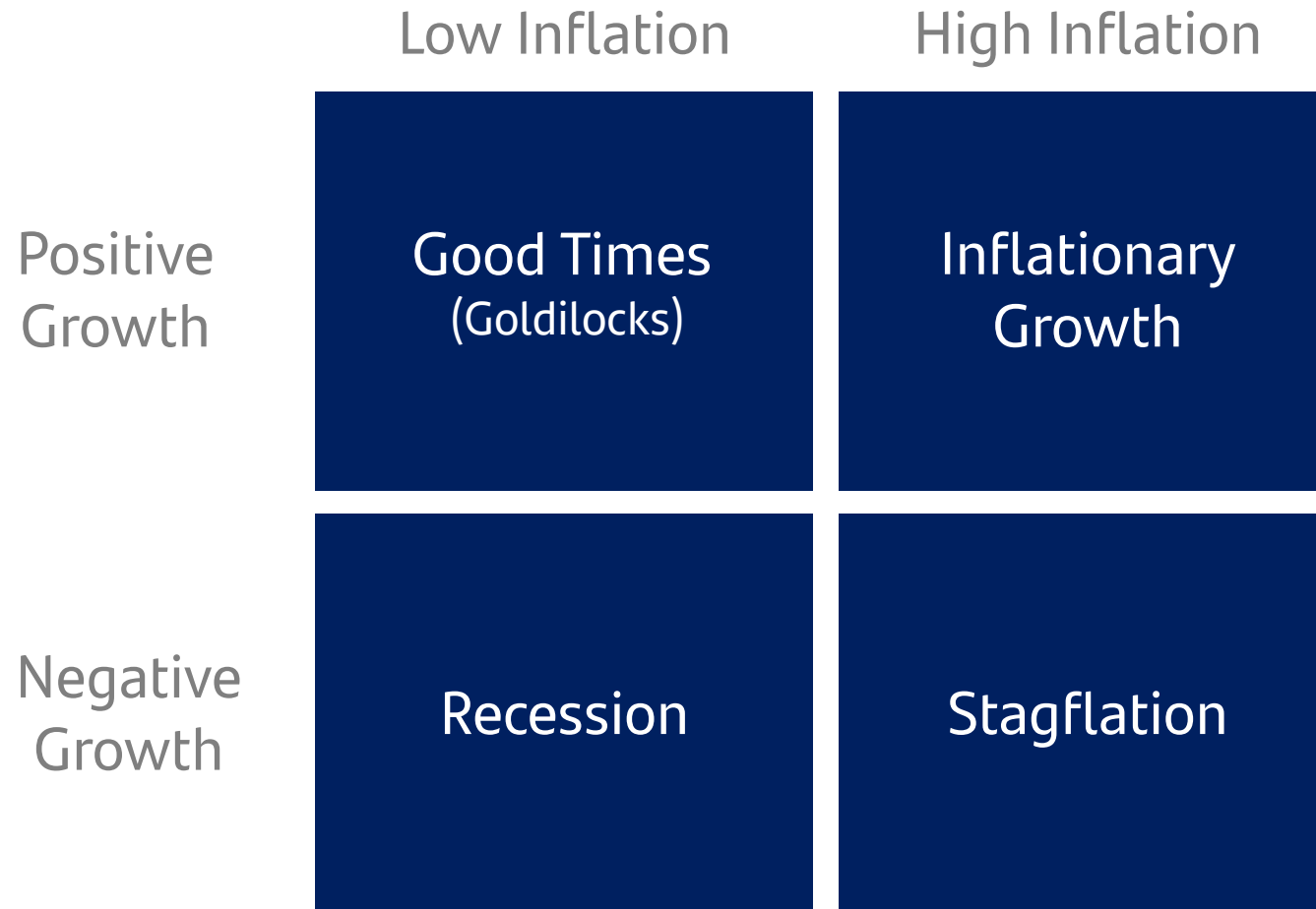
Asset Allocation Across the Business Cycle



Source: Oppenheimer Funds. U.S. monthly annualized excess returns (Jan 1970- Dec 2013).

	Contraction	Recovery	Expansion	Slowdown
Markets	Change to "Risk Off"	Change to "Risk On"	Consolidated Up-Trend	Range Bound + Dispersion
Expected Returns	Negative for risk assets	Above historical average	Better for risk assets	Close to risk free rates
Asset Classes	Fixed Income "Gobies"	Equities IG Corp & HY	Equities Commodities	Money Market
Sectors	Cons. Staples, Pharma	Financials, Industrials	Technology, Real State	Basic Materials, Energy
Styles	Low vol Quality	Value High Beta	Value Momentum	Growth Quality
Absolute Return	Global Macro	Distressed Secs, Relat. Value, Deep Value	Special Sits., L/S Equity	Merger Arb. Capital Struct. Arb.
Optimal Risk Profile	Conservative	Moderate	Agressive	Prudent

Asset Allocation Across the Business Cycle



Objectives



1

Determining Growth Trends in different Economic Regions

2

Infer Cycle stage based on these indicators

3

Set Optimal Asset Allocation based on the stage of the cycle

4

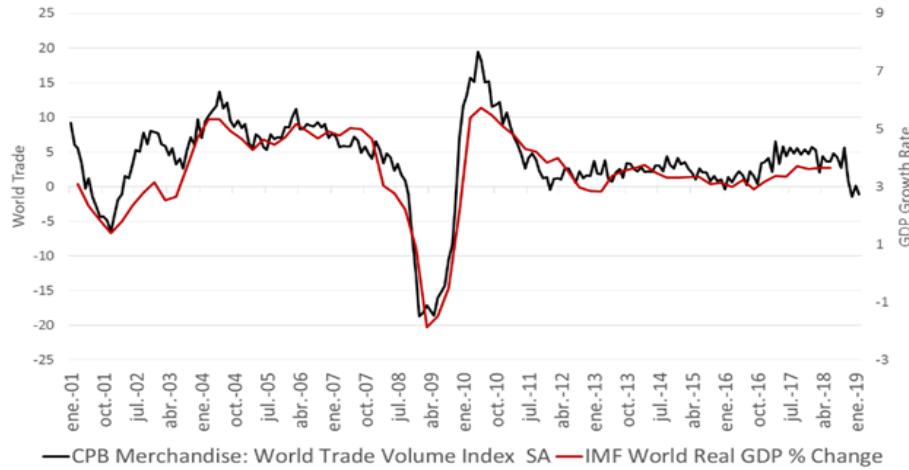
Contrast macro behavior with Market behavior to confirm we are in the correct regime, or find new market opportunities

What we believe drive Market Returns: Regimes of Growth and Inflation

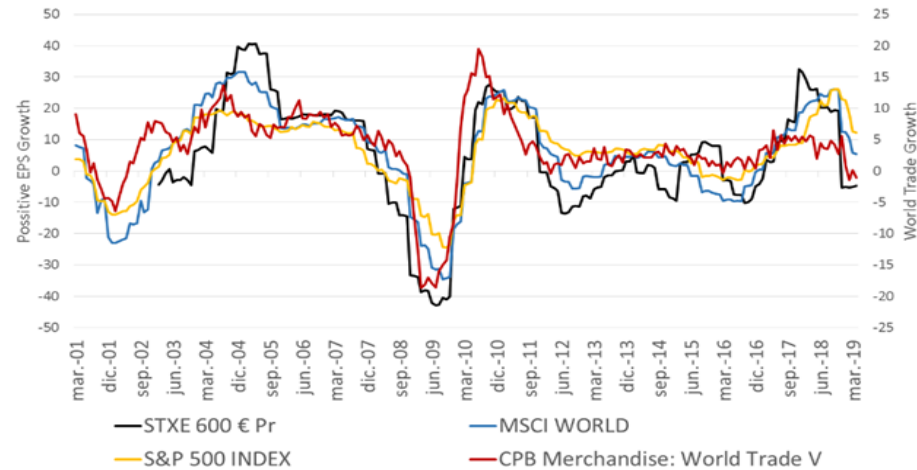
Linked Economies

- Global activity growth is what determines growth in expected Earnings growth, which in turn this determines the Investment and Business Cycle
- Use handful of information rich indicators; that signal strong regime changes within economic activity by looking at first and second derivatives of activity
- Markets are all interconnected (e.g.. we can determine what happens in Europe by looking at China)

Determining World Growth is Key:
IMF GDP Growth Rates vs World Trade



EPS Growth is a function of Global Economic Activity
Positive EPS Growth vs World Trade

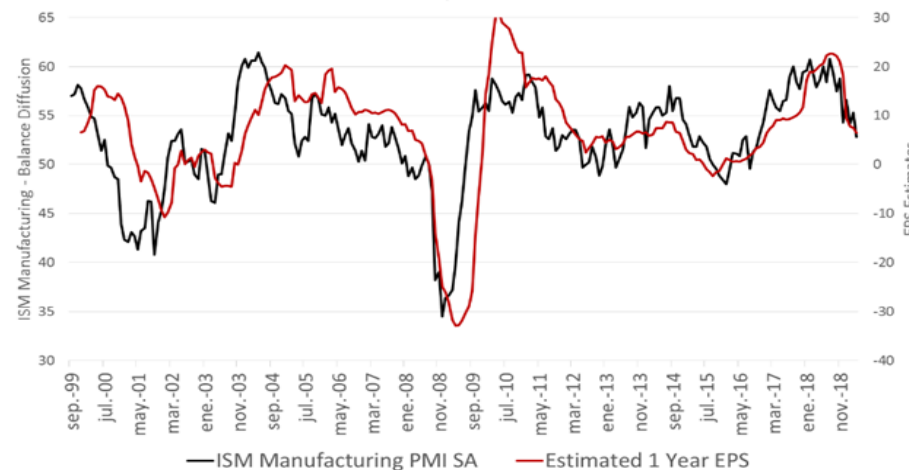


Markets are all
Interconnected...

Correlation with Global Trade

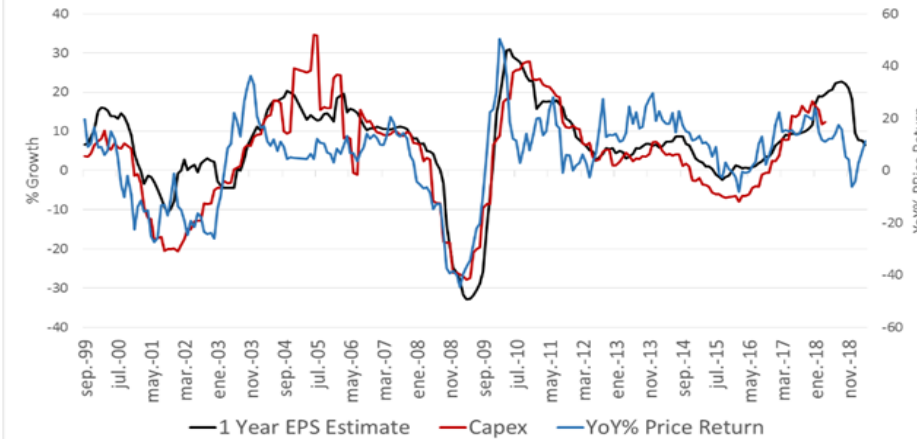
MSCI WORLD	0,70
S&P 500 INDEX	0,63
STXE 600 € Pr	0,75

The New Relevant Macro Theme is US Investment Cycle Rolling Over
US ISM Manufacturing and 1 Year EPS Estimates



Market Expectations on Earnings and Growth Lead the Investment
Cycle and Market Returns

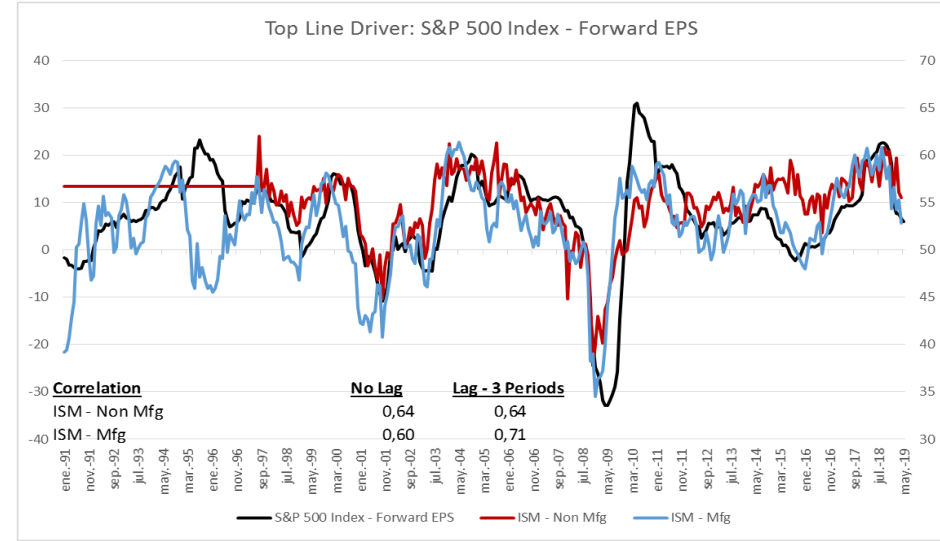
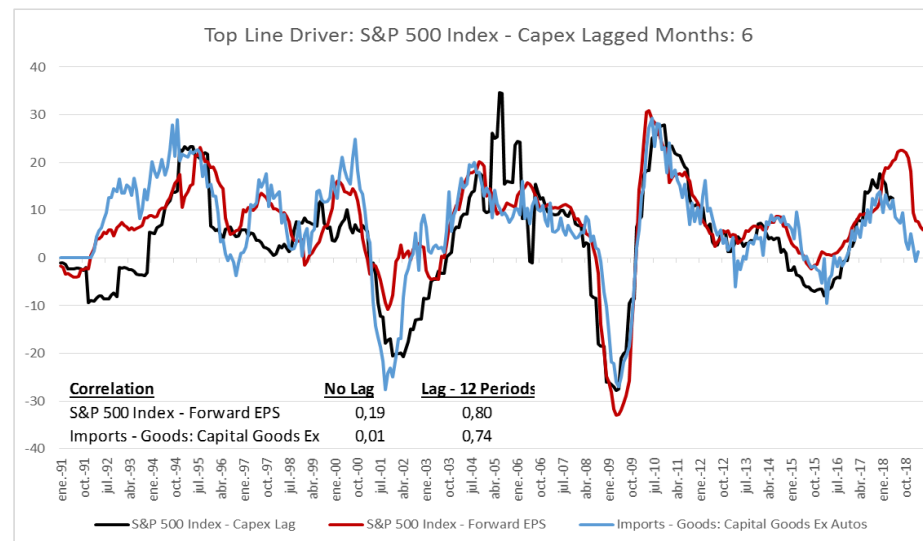
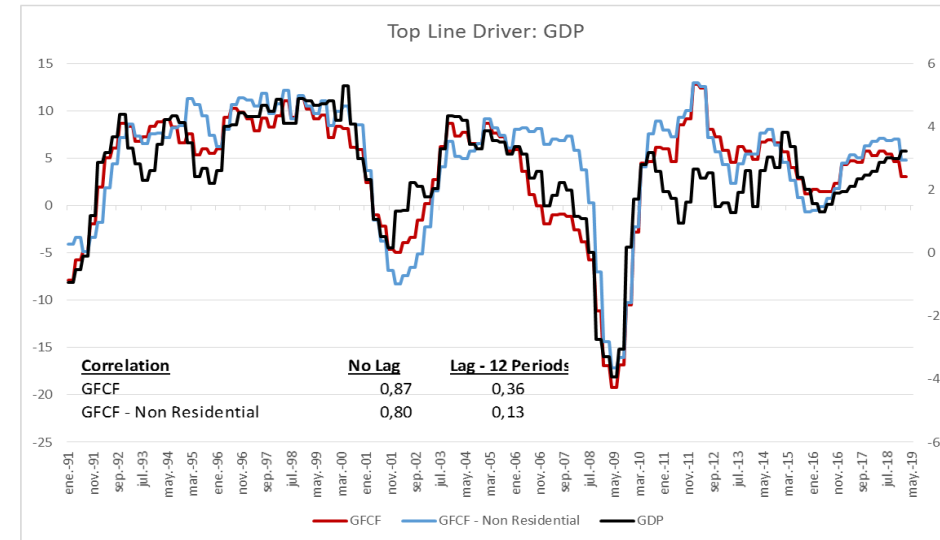
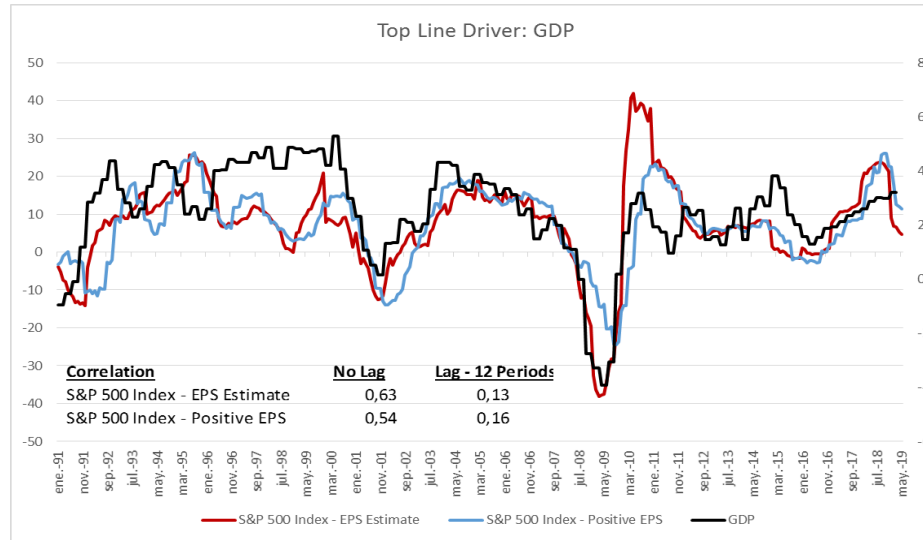
1 Year EPS Estimates and Capex (lagged 1 year) vs S&P Yearly Return



The Investment & Inventory Cycles are Key: US example on how we determine the drivers of market returns

Investment Cycle is Key

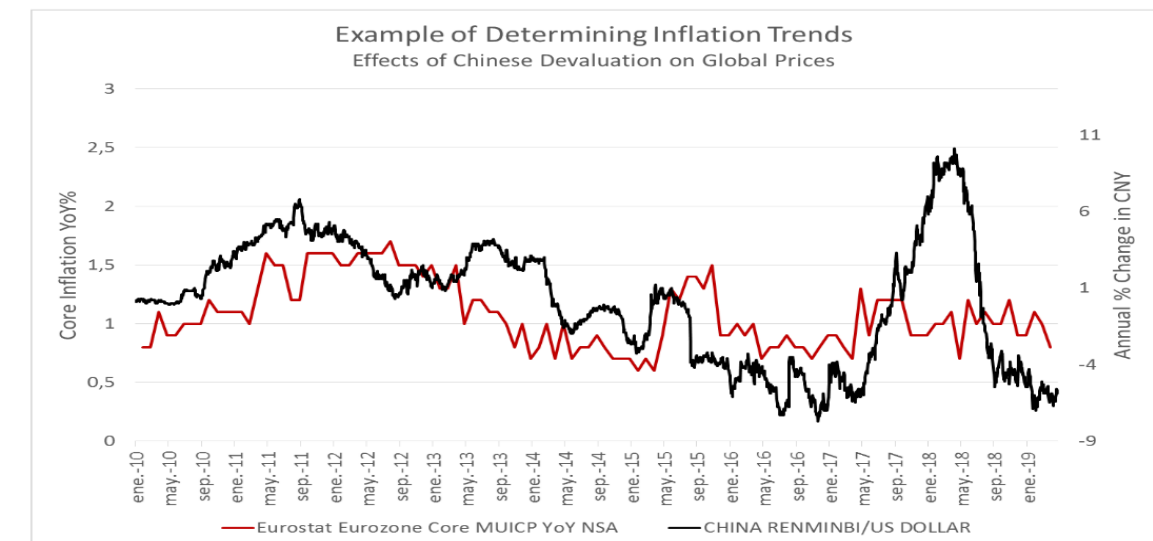
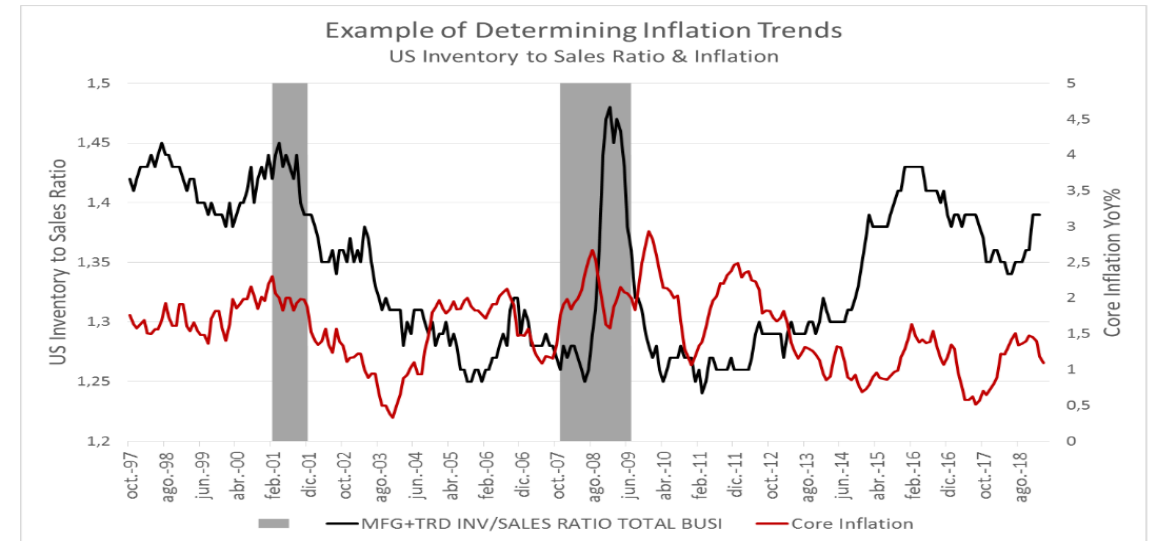
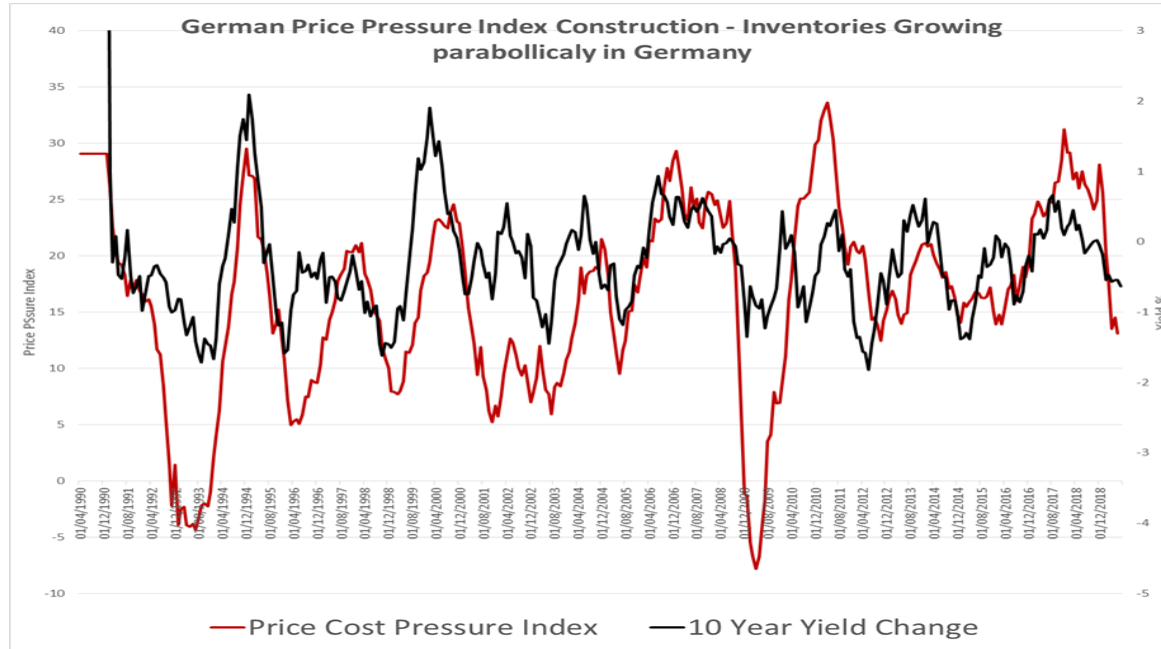
- Earnings growth and Expected Earning Growth are a function of overall economic activity growth
- The Investment & Inventory cycle determine what the prospect for growth are in the overall economy
- We focus on “high beta” indicators of the Investment Cycle (e.g. Imports of Capital Goods, ISM Mfg, etc.)

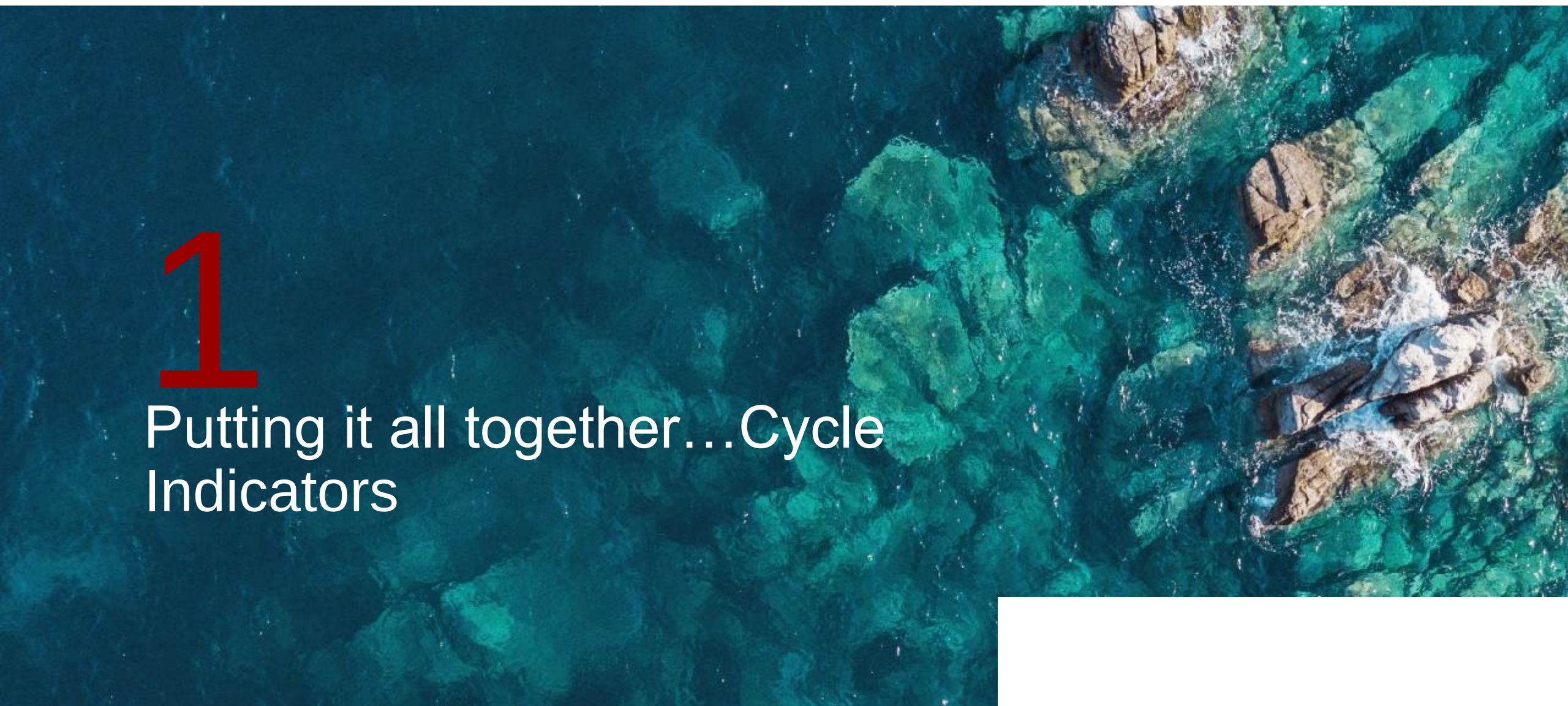


Inflation; the second most important Driver for markets Return

How we interpret inflation?

- Inflation traditionally has been a function of monetary policy and growth in monetary aggregates, but these relations are less evident as a consequence of the decline in the velocity of money
- We focus on actual sale prices expectations, inventory levels and the effect of the Chinese yuan to determine **Price Cost Pressure Indices** instead

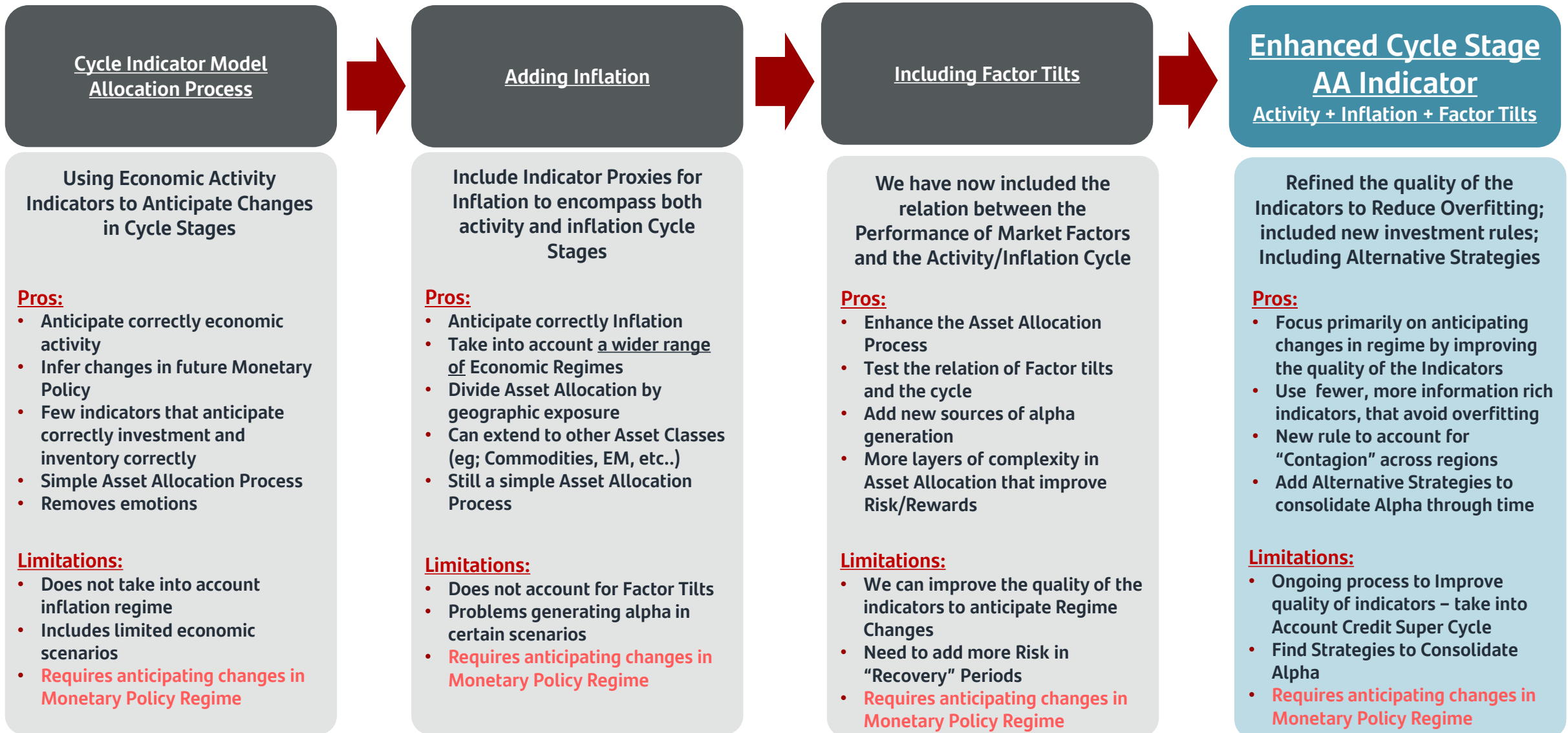


An aerial photograph of a rugged coastline. The water is a deep, dark teal on the left, transitioning to a lighter, vibrant turquoise on the right where it meets a rocky shore. The rocks are brown and jagged, with white foam from the waves crashing against them. The overall scene is dynamic and natural.

1

Putting it all together...Cycle
Indicators

Evolution of Cycle Stage Asset Allocation Process



Determining the Stage of Cycle

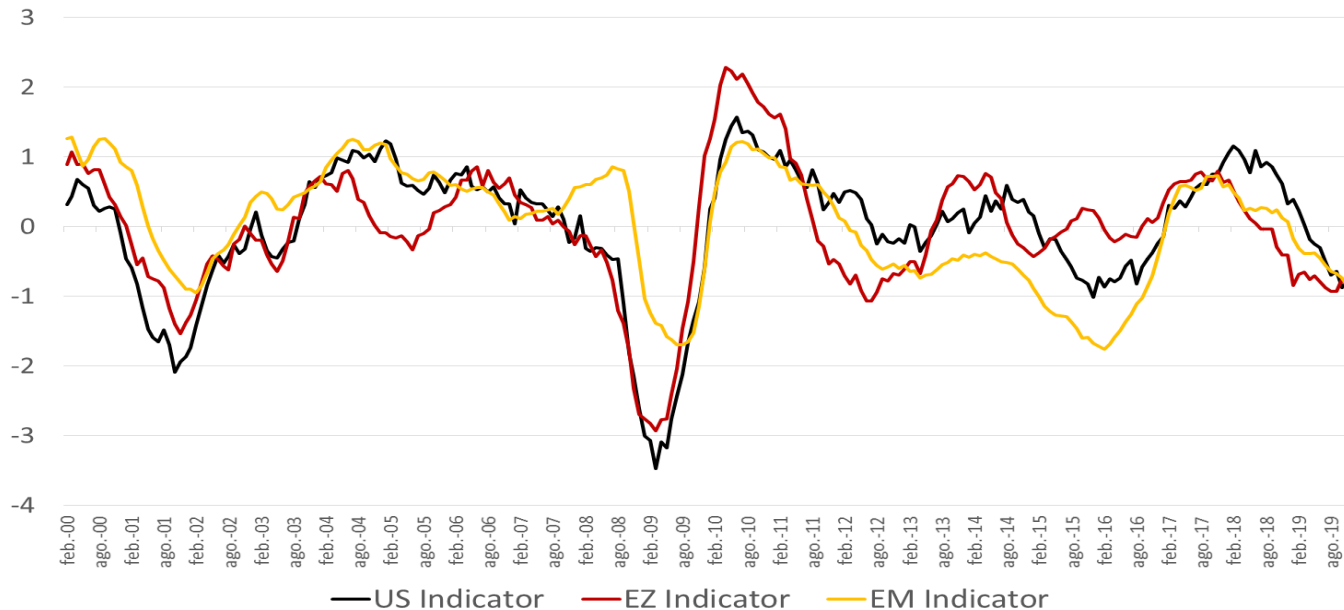
Leading Indicators

- Published quickly
- Use handful of information rich indicators; that signal strong regime changes within economic activity looking at first and second derivatives of activity
- Divide into 4 Quadrants of Business Cycle Stage (Expansion, Recession, etc.)

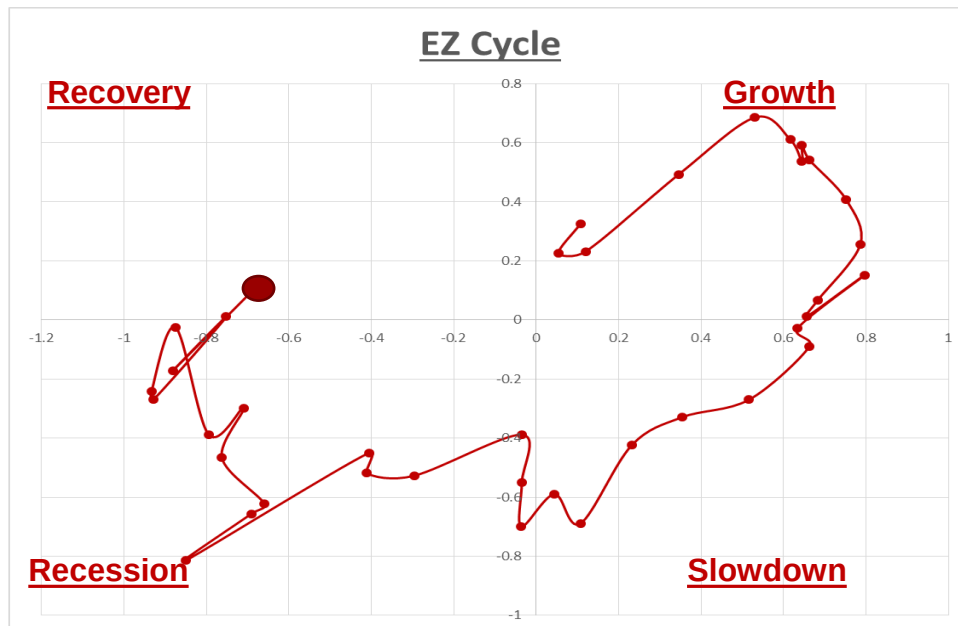
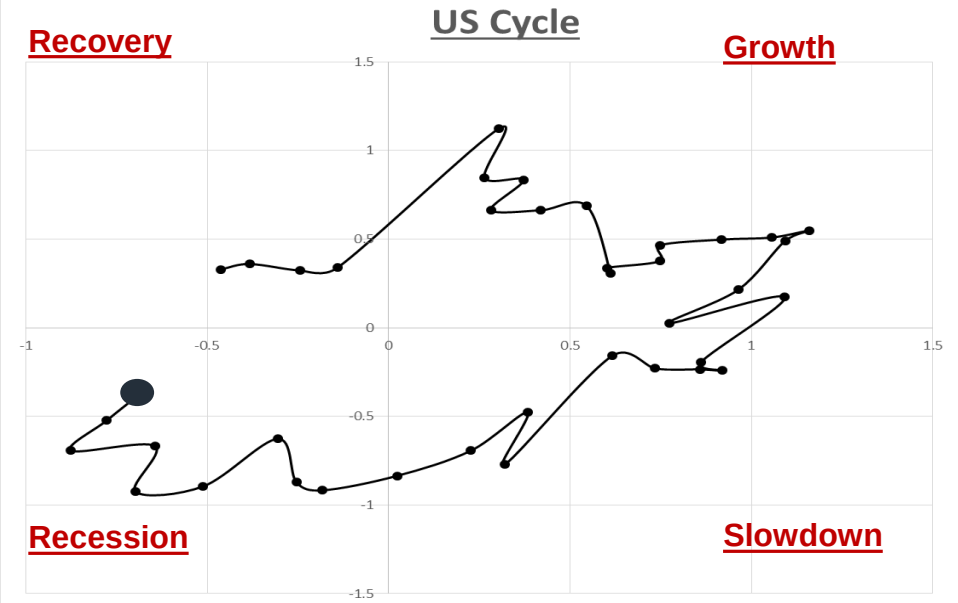
Criteria for Selection

- Correlates/anticipates correctly the following:
 - Stocks vs bonds returns
 - Real Trade Growth and global economic activity
 - Forward EPS Growth
 - Investment/Inventory Cycle

Evolution of Activity Indicators throughout the Cycle



Dividing Indicators into "Cycle Stages"



Accounting for Inflation Dynamics

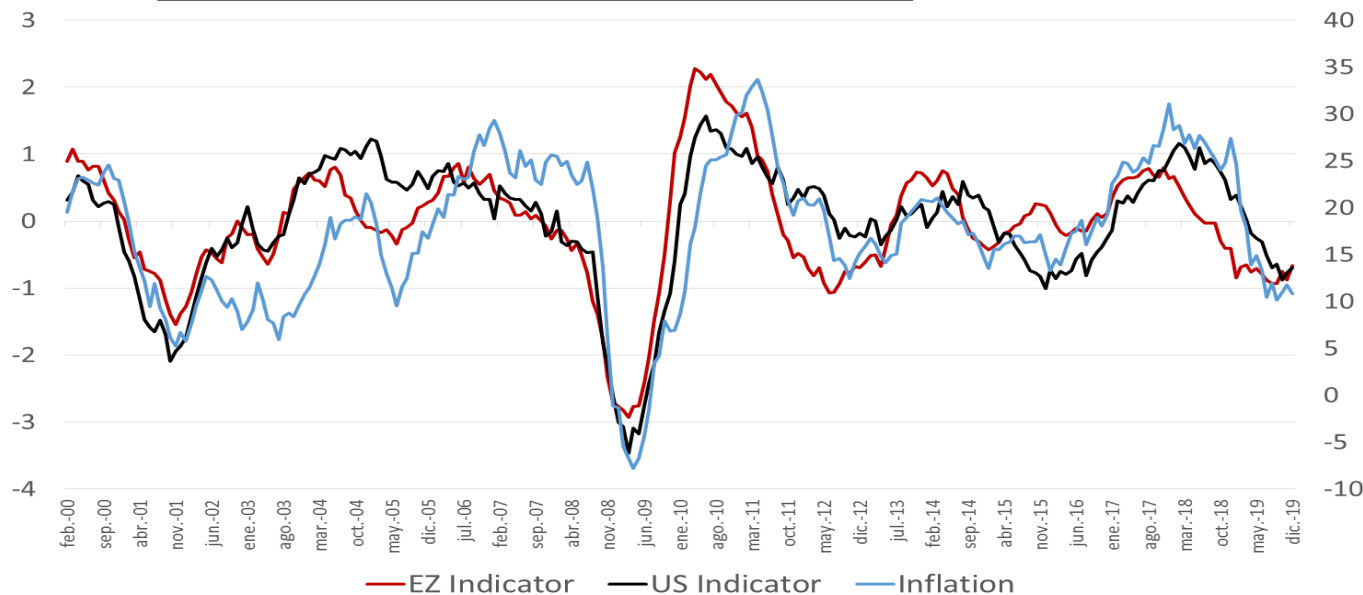
Anticipating Inflation

- Creating composite Indicators of Inflation
- Based on survey data we capture the underlying Cost Price Pressure
- Taking into account inventory levels, expected sales, and inflation expectations

Combining all the Signals

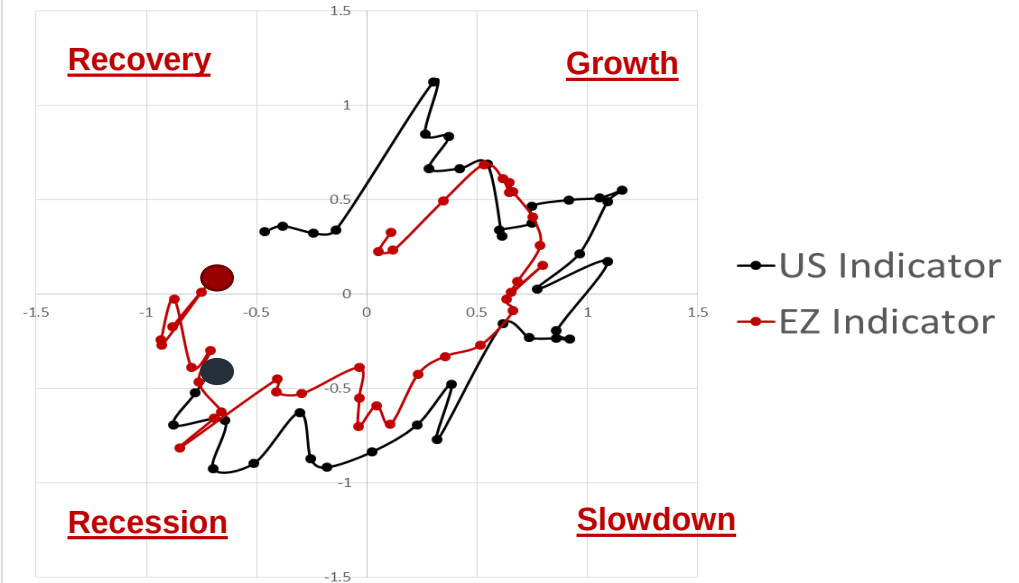
- Divide all available info into "Activity" and "Inflation" Regimes
 - Account for all possible future scenarios regarding market dynamics
 - Anticipate Central Bank Activity to quickly respond to changes in Monetary Policy and quantify its actual effect
 - Choosing the best geographic area to invest in

We have added the New Dimension of Inflation

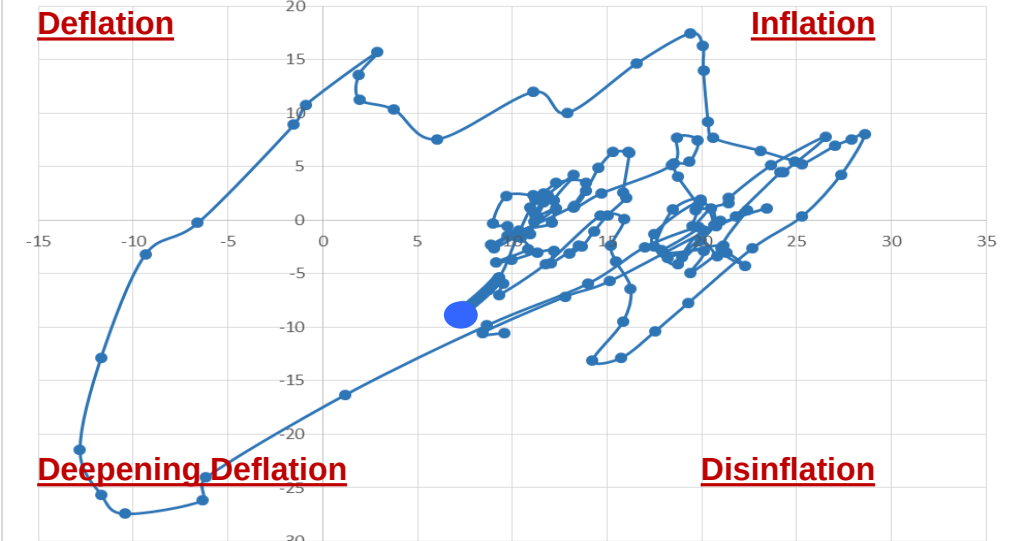


We combine Both Signals by Region

We have two Simultaneous Cycles in "Recession"



With an Inflation Overlay...



How do we Construct the Indicators?

The **Composition of the Indicators** have been Modified in a process of simplification to avoid as many Market Price components as possible, capture the underlying changes in regime in a fast manner

Composition of Indicators

US Indicator

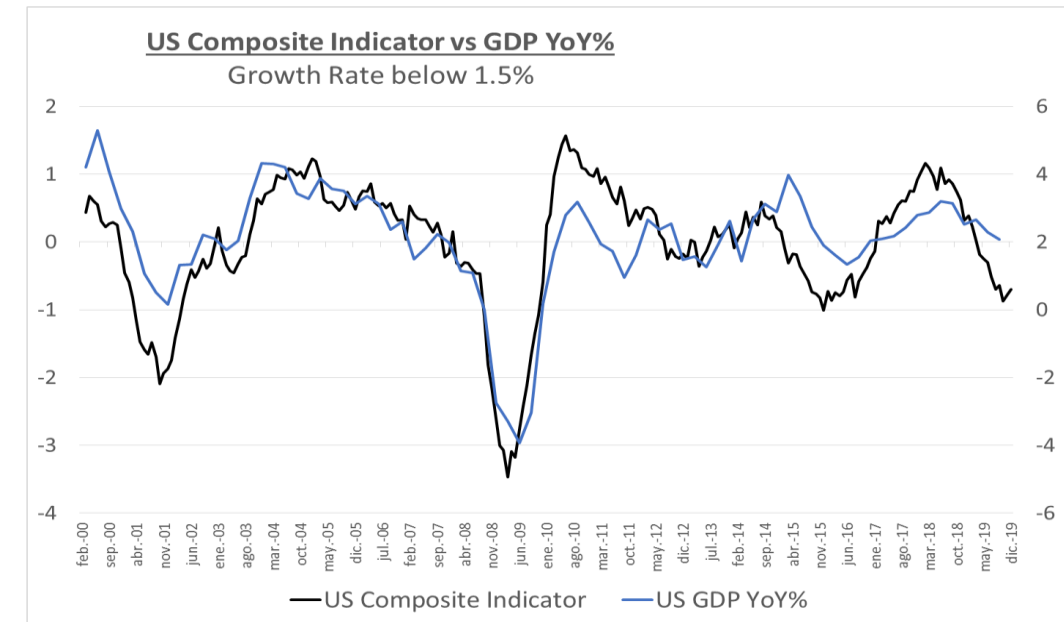
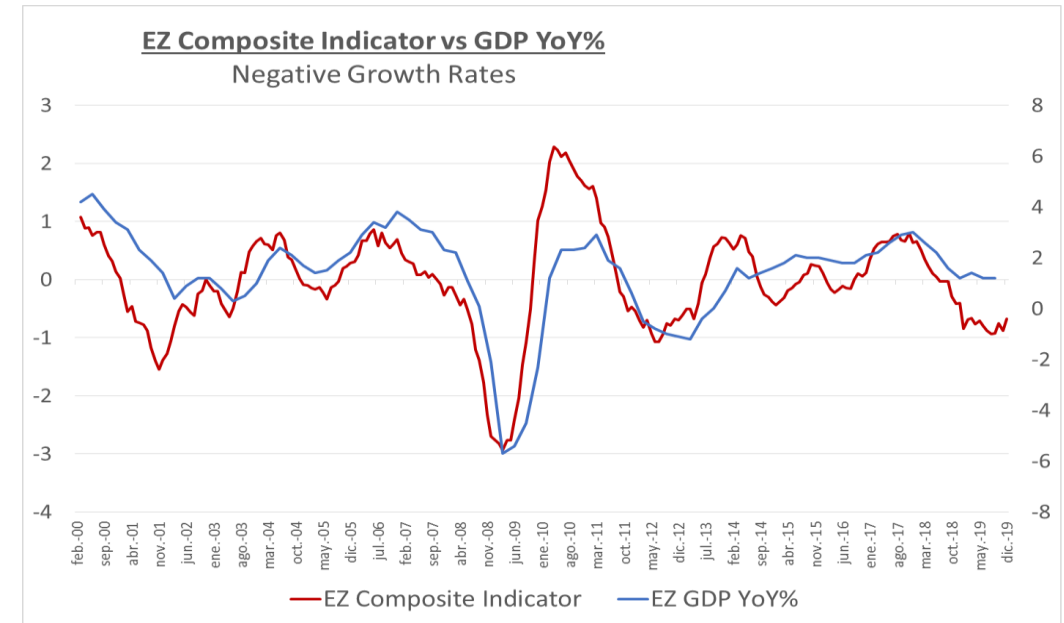
- ISM Manufacturing: Imports
- ISM Manufacturing
- US Capacity Utilization Rate
- S&P EPS Estimate 1 Year
- Core Capital Good Shipments
- Imports of Capital Goods

EZ Indicator

- EC Economic Sentiment Index
- IFO Autos
- IFO Capital Goods
- Stoxx EPS Estimates 1 Year
- EC Service Confidence

Inflation Indicator

- Construct Price Pressure Cost Index using surveys:
 - IFO
 - Philli Fed
 - NFIB
- Account for Price Expectation and Inventory Levels
- Yuan



```
graph LR; AR[Activity Regime] --> IR[Inflation Regime]; IR --> OAA[Optimal Asset Allocation]; AR --> IR; IR --> OAA; AR --> IR; IR --> OAA;
```

Activity Regime	Inflation Regime	Optimal Asset Allocation
"Growth"	"Inflation"	Stocks
"Slowdown"	"Disinflation"	Sovereign Bonds
"Recession"	"Deflation"	Investment Grade
"Recovery"	"Deepening Inflation"	High Yield

→ We would have to overweight risk assets in the portfolio

Another Example would be: "Stagflation"

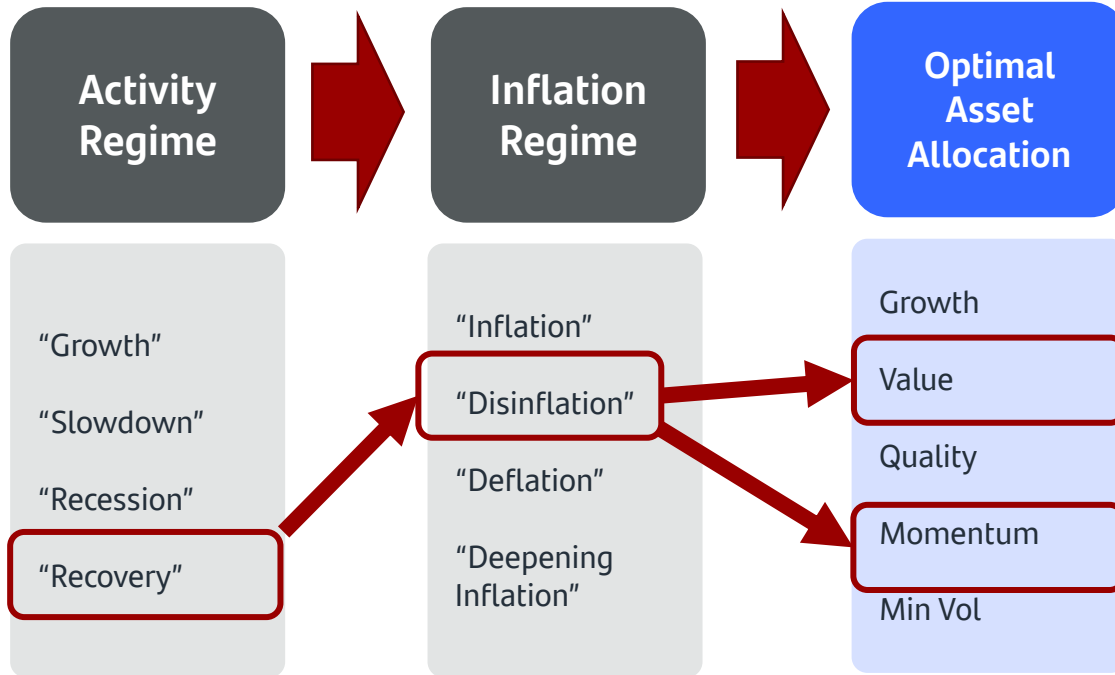
Example of Outcome and Asset Weights

Proposed Asset Allocation Grid; used for Backtest

<u>Activity</u>	<u>Inflation</u>	<u>Stocks</u>	<u>Govs</u>	<u>IG</u>	<u>HY</u>
Growth	Deepening	40	30	20	10
Growth	Deflation	30	30	30	10
Growth	Inflation	90			10
Growth	Disinflation	90			10
Recession	Deepening		100		
Recession	Deflation	30	60	10	
Recession	Inflation	30	20	30	20
Recession	Disinflation	10	70	20	
Recovery	Deepening	40	30	20	10
Recovery	Deflation	90			10
Recovery	Inflation	60		10	30
Recovery	Disinflation	80		10	10
Slowdown	Deepening	20	80		
Slowdown	Deflation	30	50	20	
Slowdown	Inflation	60		40	
Slowdown	Disinflation	40	50	10	

The Logic Behind the Cycle Stage Asset Allocation Process

Determining the Factor Tilts



Example of Outcome and Factor Tilts

Activity	Inflation	GROWTH	VALUE	QUALITY	MOMENTUM	MIN VOL	SIZE
Growth	Deepening	30	0	0	40	0	30
Growth	Deflation	0	30	0	40	0	30
Growth	Inflation	0	30	0	40	0	30
Growth	Disinflation	30	0	0	40	0	30
Recession	Deepening	0	0	50	0	50	0
Recession	Deflation	0	25	25	0	50	0
Recession	Inflation	0	25	25	0	50	0
Recession	Disinflation	0	0	50	0	50	0
Recovery	Deepening	0	40	0	30	0	30
Recovery	Deflation	0	40	0	30	0	30
Recovery	Inflation	0	40	0	30	0	30
Recovery	Disinflation	0	40	0	30	0	30
Slowdown	Deepening	30	0	30	0	40	0
Slowdown	Deflation	30	0	30	0	40	0
Slowdown	Inflation	0	0	60	0	40	0
Slowdown	Disinflation	30	0	30	0	40	0

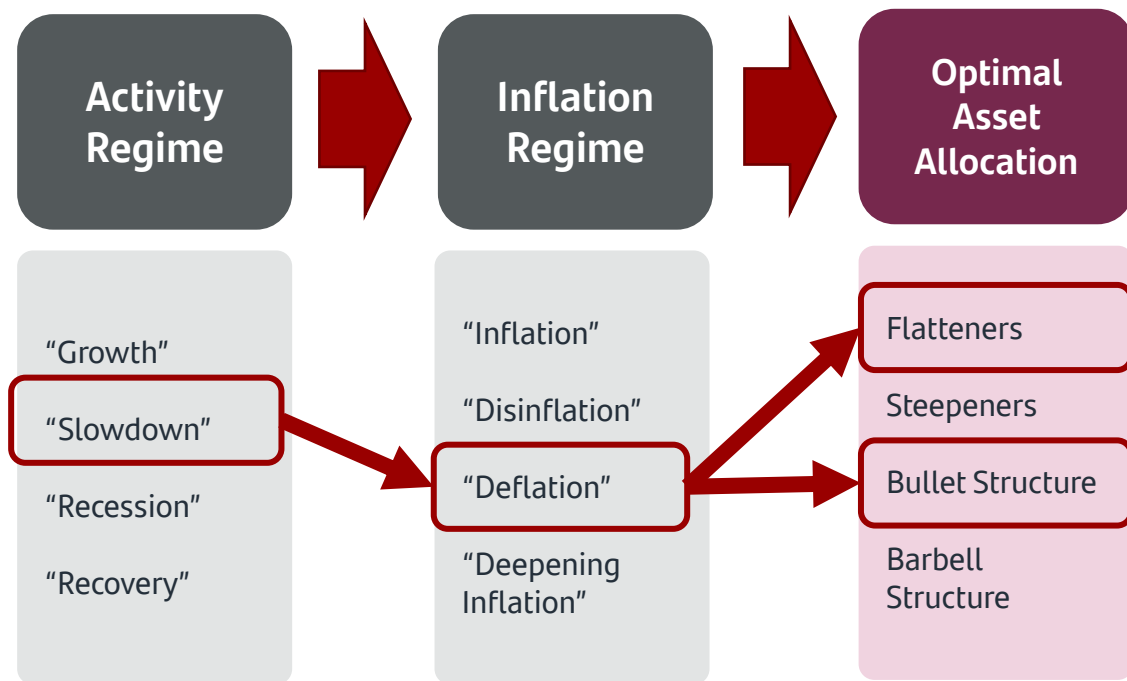
Example: Right after a big Recession where inflation is usually timid or there is even deflation...**"Post Crisis Scenario"**

→ We would have to overweight value stocks and try to capture companies that recover cyclically

Another Example would be: **"Falling Yield Environment"**

The Logic Behind the Cycle Stage Asset Allocation Process

Positioning across the Yield Curve



Example: towards the end of the cycle, as Central Banks tighten policy by raising rates and inflation starts to cool; ... **"Curve Flattens"**

→ We would want to add Duration Risk and bet on the Long end of the Yield Curve outperforming

Another Example would be: **"Start of New Cycle"**

Example of Outcome and Curve Position

Proposed Asset Allocation Grid; used for Backtest

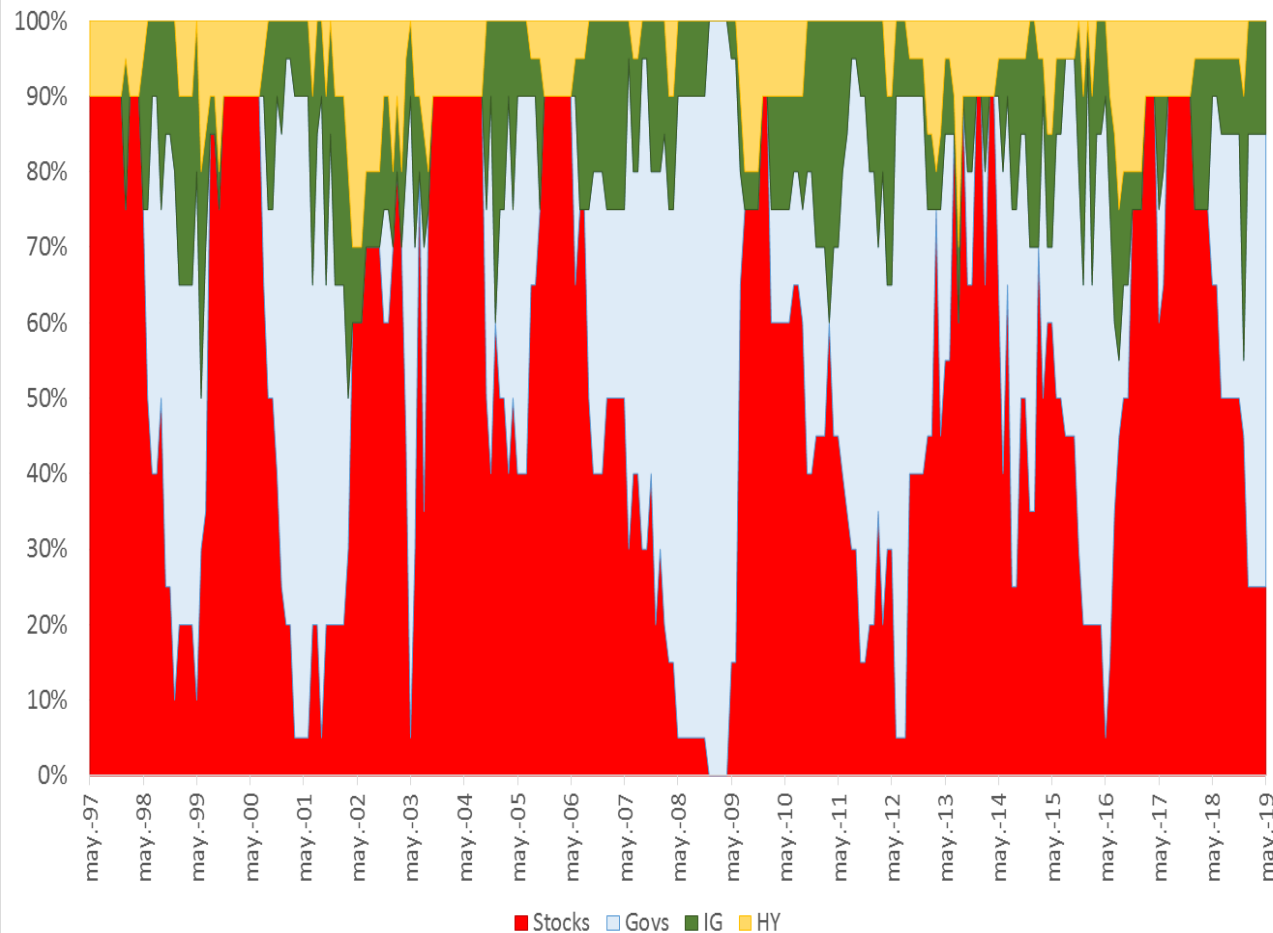
Activity	Inflation	1-3	5-10	+10
Growth	Deepening	40		60
Growth	Deflation	50	50	0
Growth	Inflation	100		
Growth	Disinflation	60		40
Recession	Deepening	40	30	30
Recession	Deflation	70	30	
Recession	Inflation	80	20	
Recession	Disinflation	70	20	10
Recovery	Deepening	60	20	20
Recovery	Deflation	70	30	0
Recovery	Inflation	80	20	
Recovery	Disinflation	70	20	10
Slowdown	Deepening	10	30	60
Slowdown	Deflation	10	70	20
Slowdown	Inflation	40	60	
Slowdown	Disinflation	20	40	40

How does all this translate into an Asset Allocation Process?

Evolution of Portfolio Composition throughout the Cycle – Turning before Market Regime changes

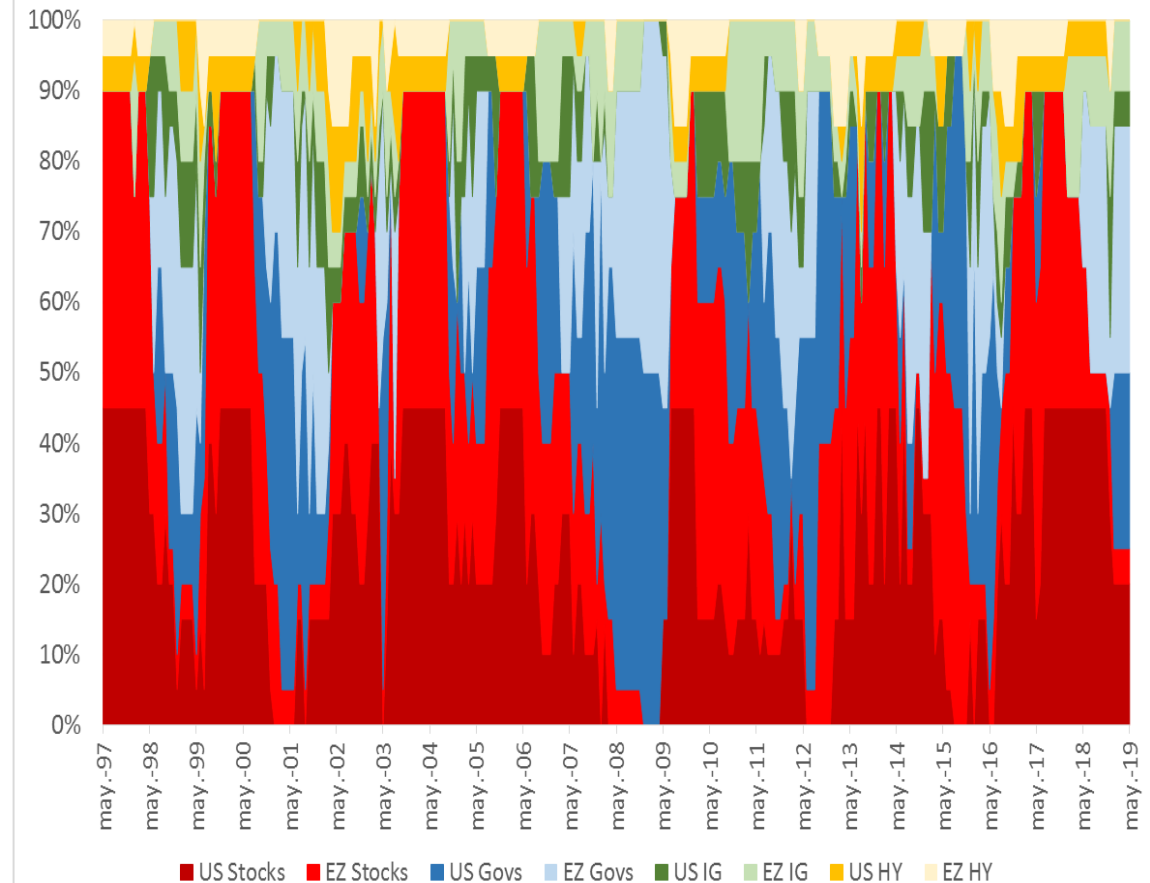
Evolution of Portfolio Asset Allocation:

Dynamically Changes with Cycle



Evolution of Portfolio Asset Allocation:

Among Best Geographic Area

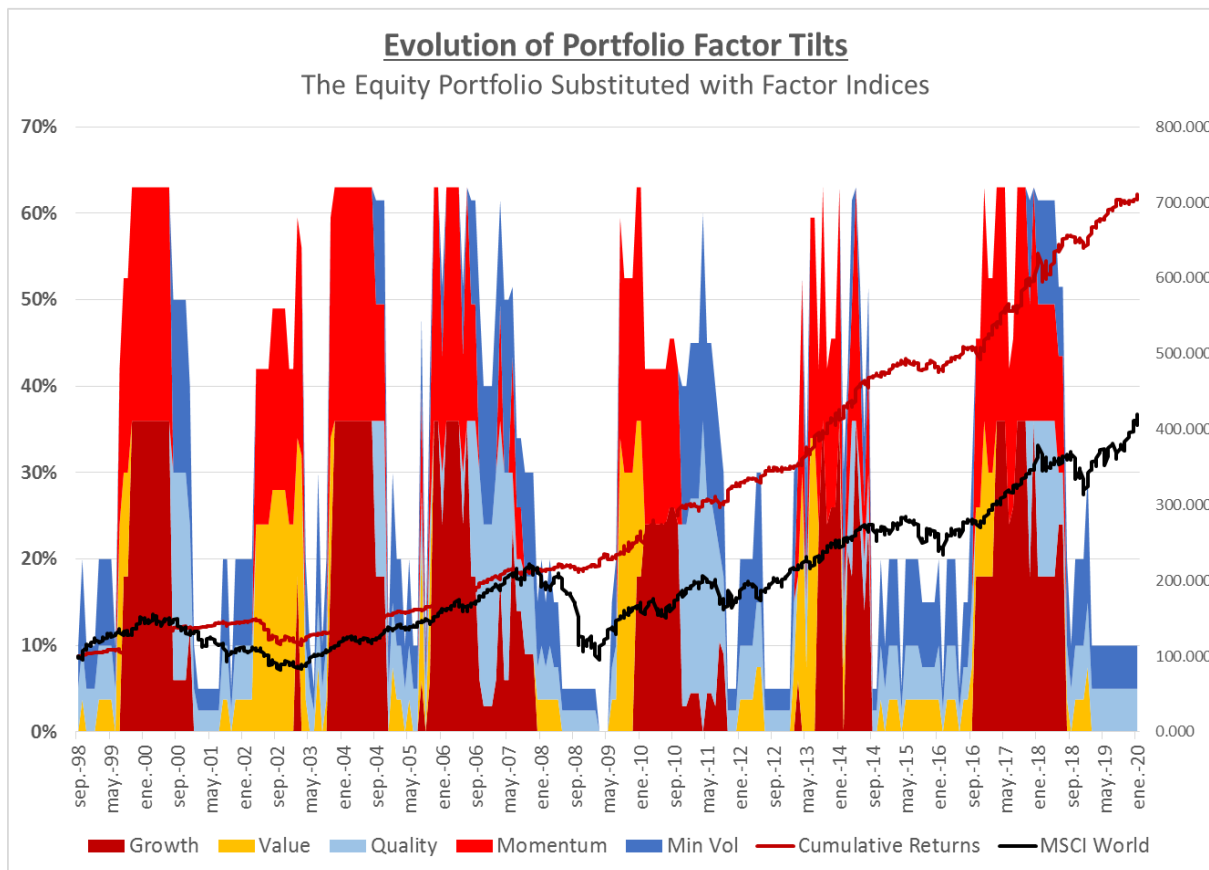


How can we capture more Alpha Across the Cycle?

Evolution of Portfolio Composition throughout the Cycle – Enhancing Cumulative Alpha with Factor Tilts

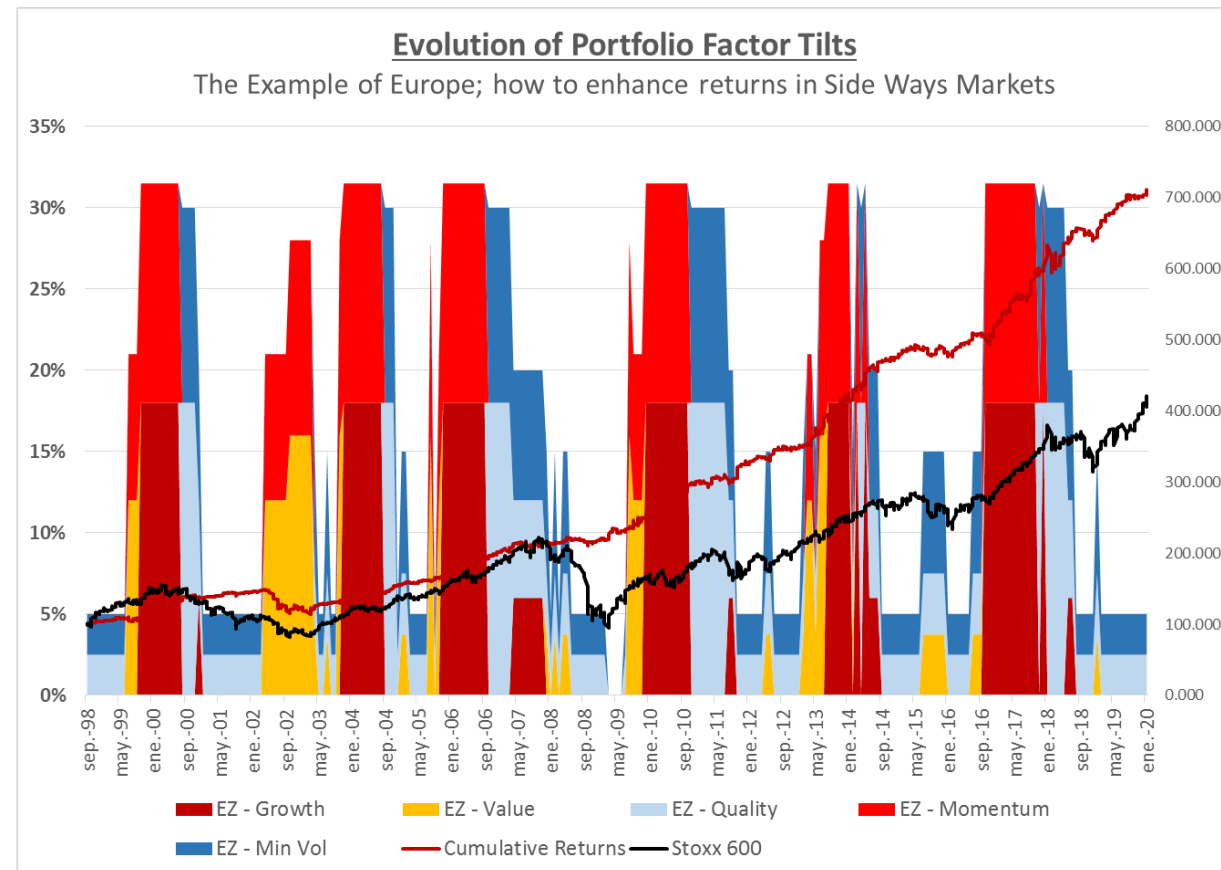
Evolution of Portfolio Factor Tilts

The Equity Portfolio Substituted with Factor Indices



Evolution of Portfolio Factor Tilts

The Example of Europe; how to enhance returns in Side Ways Markets

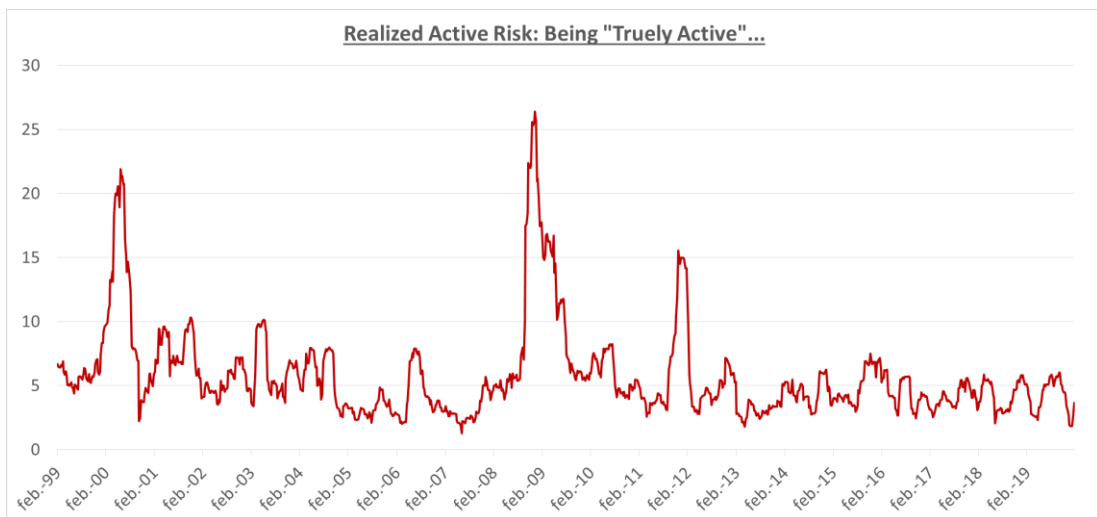


What are the Results?...Try to be "Truly Active"

Taking Active Risk when Needed

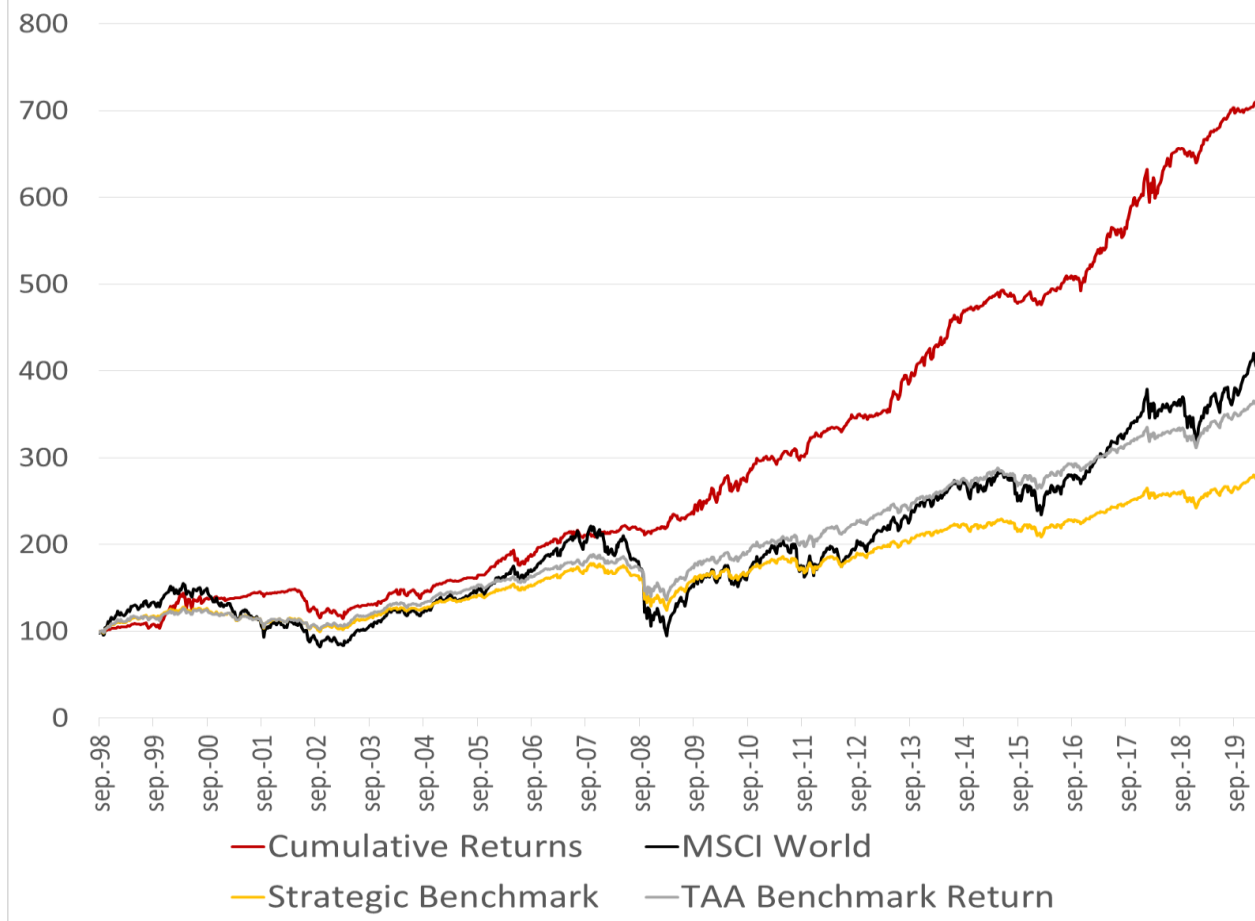
Risk Metrics

	Cumulative Returns	TAA Benchmark	MSCI World
Annualized Return	9.60	4.73	3.50
SD of Yearly Return	5.45	8.60	16.57
Risk Adjusted Return	1.76	0.55	0.21
Max DD	-8.20	-30.33	-58.37
Calmar Ratio	1.17	0.16	0.06
Max Weekly Loss	-5.49	-10.78	-20.05
Max Weekly Gain	3.60	6.98	12.34
Annualized Alpha	4.87		
Active Risk	8.15		
Information Ratio	59.74%		



Back tested Results Look Promising

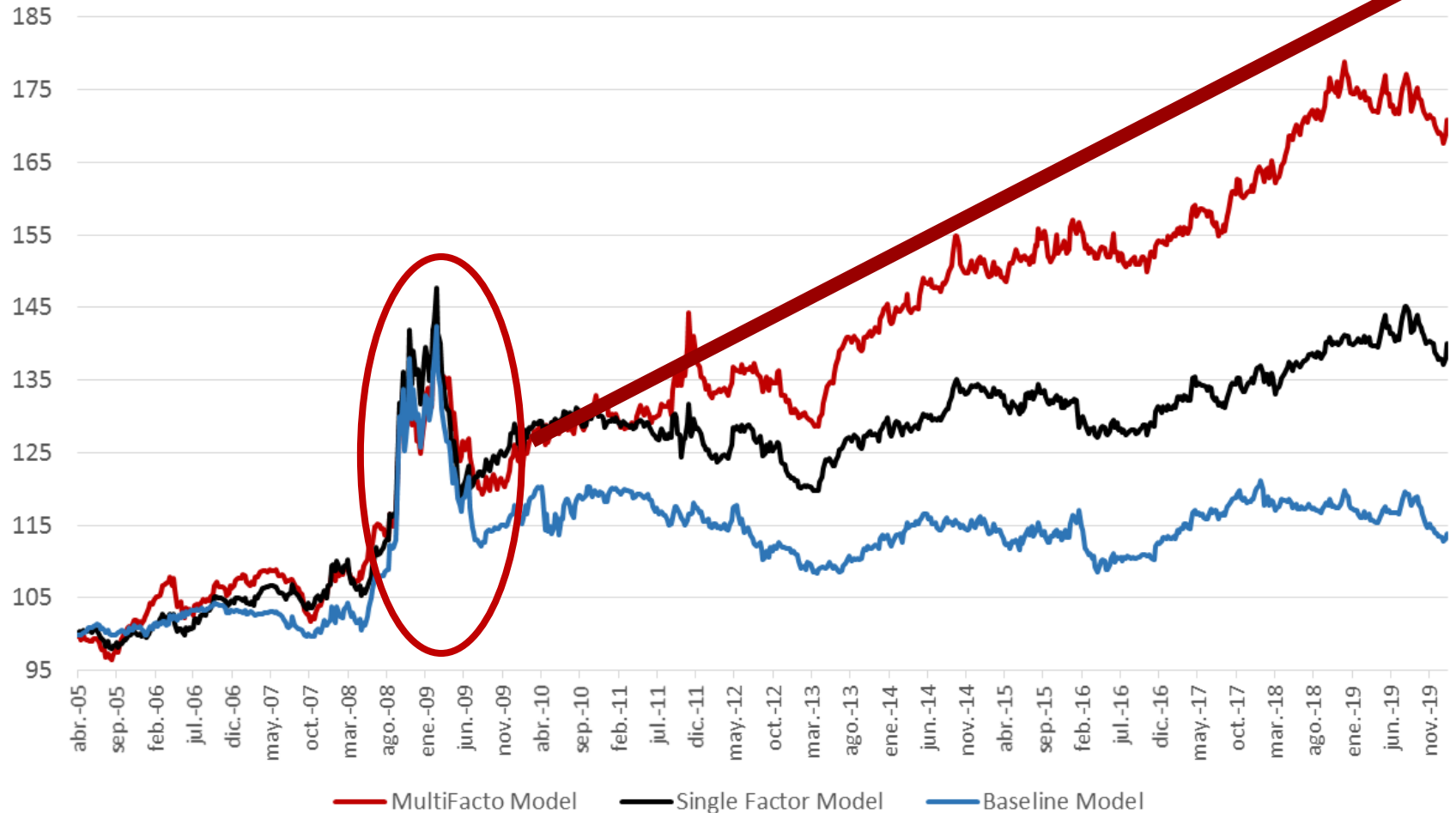
Backtested Results of New Methodology TAA Benchmark: Portfolio of 50/50 Stocks and Bonds



*The QIS Benchmark shows higher returns given than the testing period only starts from 2004. Back test of Current Asset Allocation Process from 2004 shows improved results.

Have these changes in Methodology worked...? How we have improved results from the original Model

Cumulative Alpha - Model Evolution



Capturing Alpha

The Biggest Limitation for this Investment Process is the inability to include systematic methods to account for Changes in Both Fiscal and Monetary Policy

The biggest Challenge is Monetizing Alpha in Regime Changes and Recovery Periods

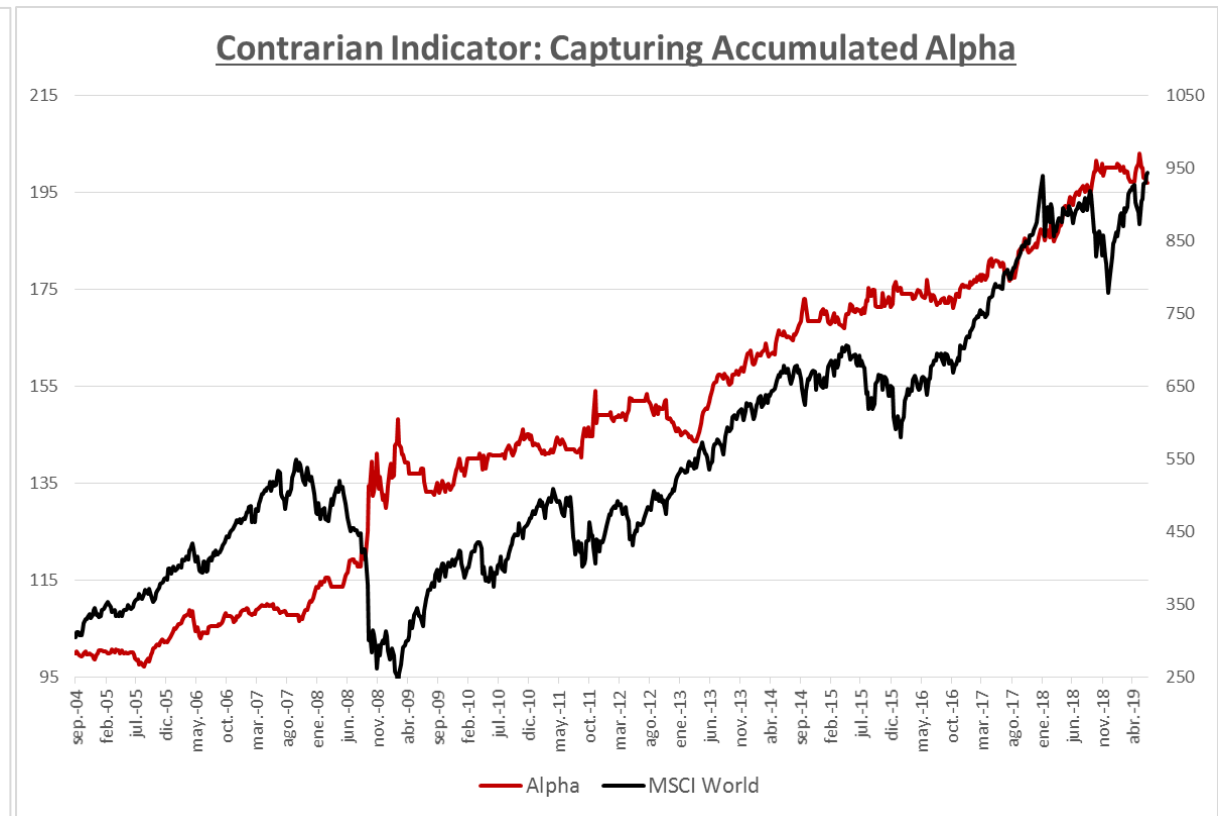
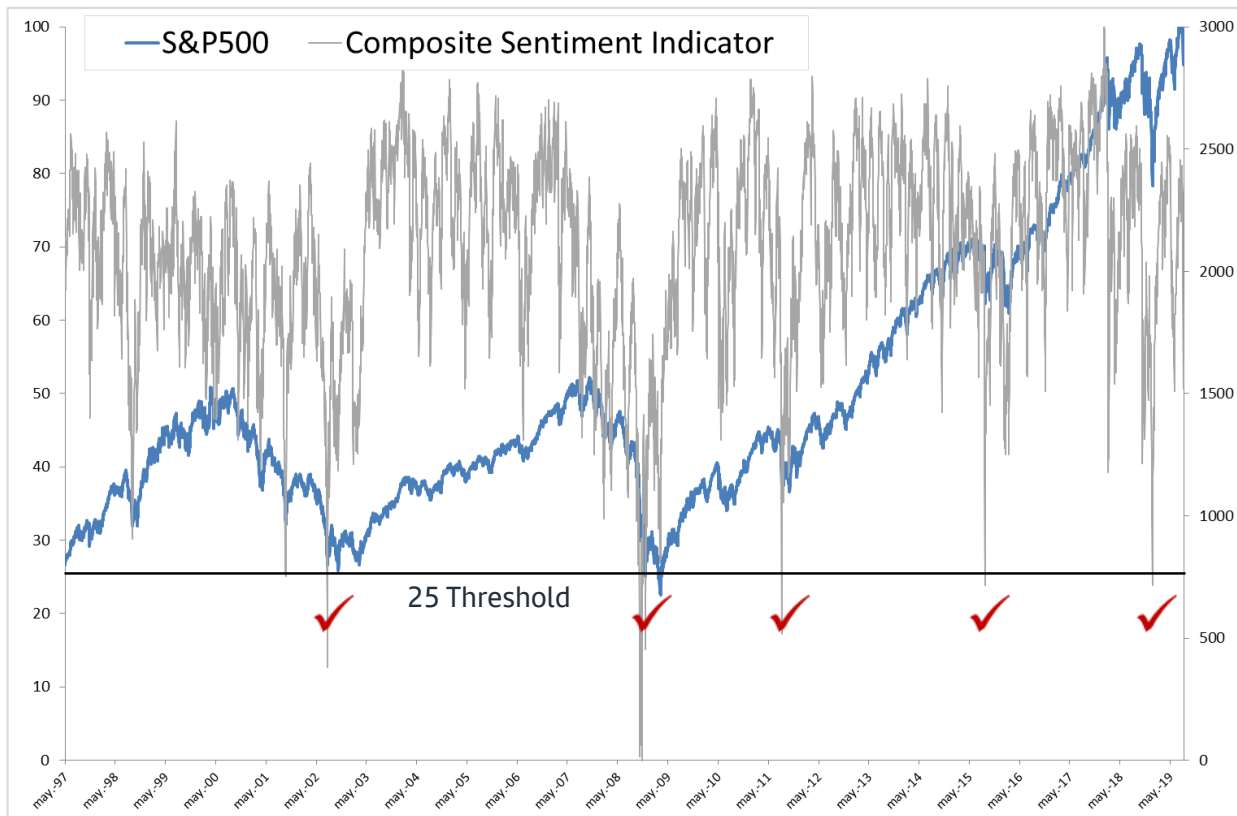
→ Alternatives:

- Option Strategies
- Monetary Policy "Triggers" to increase Exposure
- Credit Super Cycle Inclusion

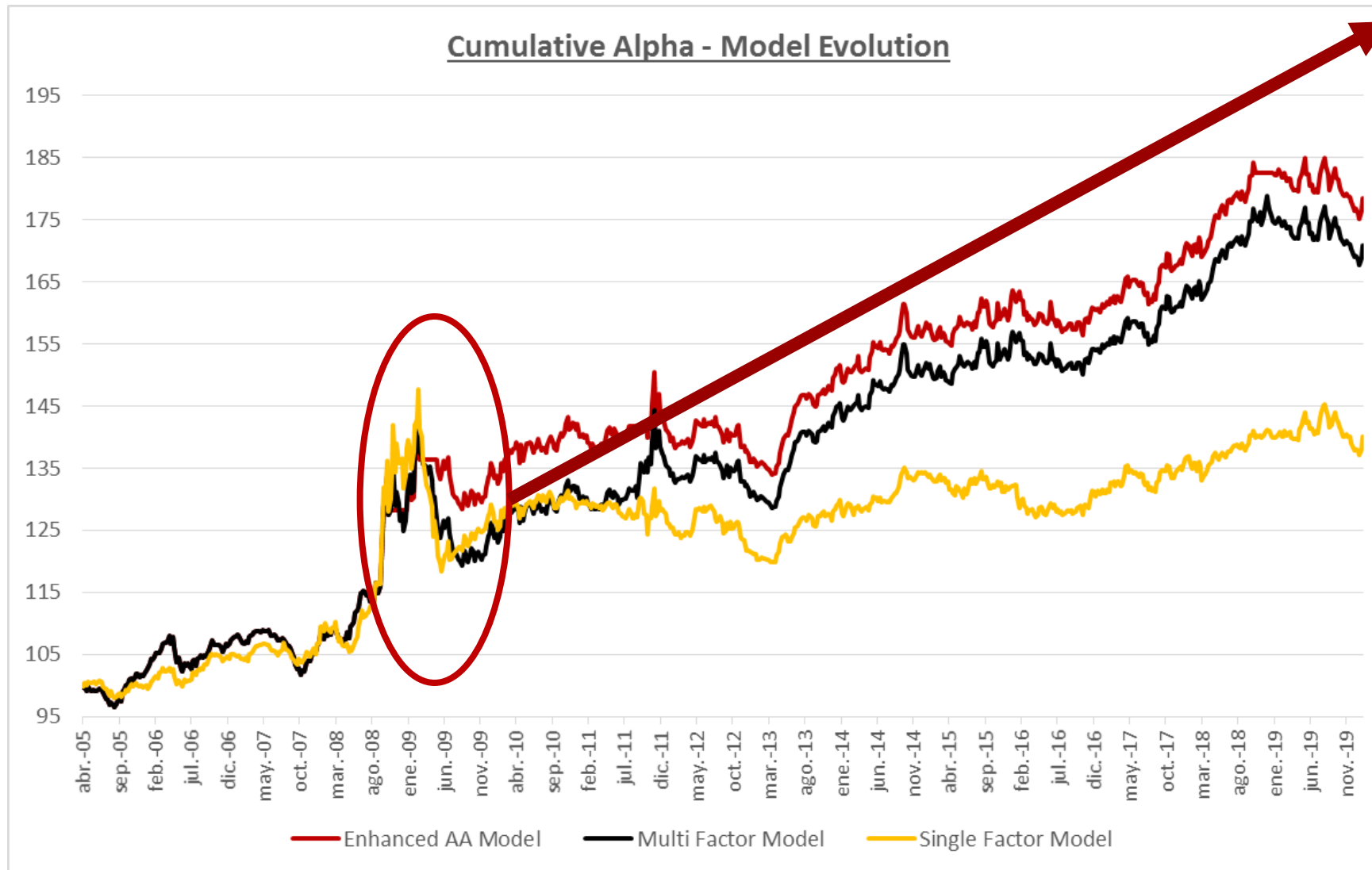
How have we corrected the issue of losing Alpha through Monetary Policy Triggers?

Contrarian Indicator

We have constructed a Composite Sentiment Indicator, to help us capture periods of extreme negativity and take those levels to Neutralize the Portfolio → **Avoid Losing Cumulative Alpha**



Have these changes in Methodology worked...? How we have improved results from the original Model



Capturing Alpha

With this Rule in our process, we have managed to take profits in our positioning, hence ensure cumulative Alpha

Introducing Momentum

- Work in Progress to improve Indicator:
 1. Smooth data to avoid Type II errors. Use the principle of statistical test to infer strength of signal, avoid false changes in regime
 - a) Smooth Data
 - b) Overlay with a less volatile trend indicator
 - c) Incorporate Rule of Regime Change; Statistical Inference to check strength of Signal Change
 2. We have Created a Peripheral Indicator as well as an Emerging Market Indicator
 3. Add a “Monetary Policy Trigger” Signal to account for changes in Monetary/ Fiscal Policy Change
 4. Include **Credit Super Cycle Regime**



2

Introducing Momentum...

Introducing Cross-Asset Momentum

The issues with the Cycle Indicator Model:

- Economic Data **does not react** in a timely manner to “Market Recoveries”
- Government and Central Bank action interferes in markets and the effect of those cannot be quantified beforehand (we cannot back test policy triggers beforehand)
- We are exposed to any market shocks that are not captured by economic indicators

How can we smooth this?

Objective

Counteract the issues with our baseline Model by introducing asset price momentum.

Strategy

- 1.- Pick the subset of assets that have the highest **Time Weighted Price Momentum** across a broad range of time periods
- 2.- Adjust by “**Quality**” of momentum to then select a narrower subset of assets.
- 3.- Introduce a trigger of when to “**Turn Off**” the strategy based on cross asset dispersion, to avoid drawdowns.

Execution

The strategy aims to **run in parallel to our baseline Model** assigning time varying weights based on the **Correlation** of both strategies.

Benefits

React more promptly to changes in Market Regimes (particularly in the “Recovery” Phases) by assigning more risk to the overall portfolio

Introducing Cross-Asset Momentum

Detailing the Process

Calculate Price Momentum

We calculate **Price Momentum** Across Different time Frames

- 3 Month
- 6 Month
- 9 Month
- 12 Month

We calculate the **number of consecutive weeks** with positive returns for each Asset

Calculate a Weighted Price Momentum

We take a **Weighted Average of the Price Momentum** across the previous Time Frames

→ **Rationale:** We **overweigh most recent time frames as we believe it contains more recent information**

Obtain the top Asset Class Subset

After we filter the **top 10 Asset Classes** with highest Weighted Price Momentum

→ **Rationale:** The idea is to favor assets which are quickly gaining momentum

Adjust for "Quality" of Momentum

From the previous subset we finally pick the **top 4 Assets** with highest number of consecutive positive weeks

→ **Rationale:** We favor **asset classes with more consistency in momentum**

When to "Turn off" the Strategy

Momentum Strategies display long term superior performance but are subject to period of **sudden drawdown**.

We apply filters to trigger a signal to **"turn off"** the strategy based on:

- Cross Asset Dispersion and Vol
- Cross-Correlation

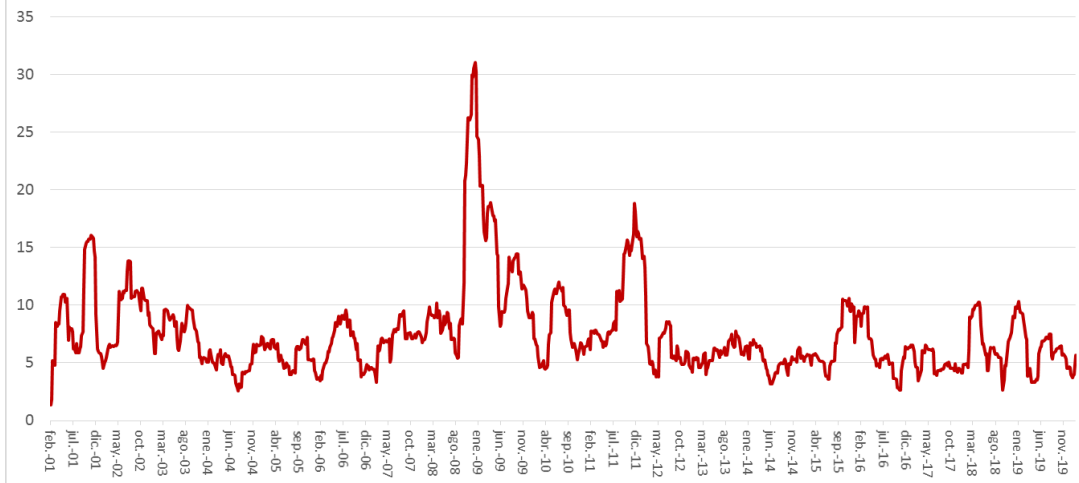
When the Strategy is **"Turned off"** the model goes into **Bonds**.

The Results of the Momentum Model tries to work in Parallel

When "Following the Herd" counts

<u>Risk Metrics</u>	Cumulative Returns	TAA Benchmark	MSCI World
Annualized Return	10.92	4.83	3.51
SD of Yearly Return	7.90	8.66	16.68
Risk Adjusted Return	1.38	0.56	0.21
Max DD	-12.86	-30.29	-58.76
Calmar Ratio	0.85	0.16	0.06
Max Weekly Loss	-5.87	-10.79	-20.07
Max Weekly Gain	4.95	6.99	12.37
Annualized Alpha	5.44		
Active Risk	8.77		
Information Ratio	62.00%		

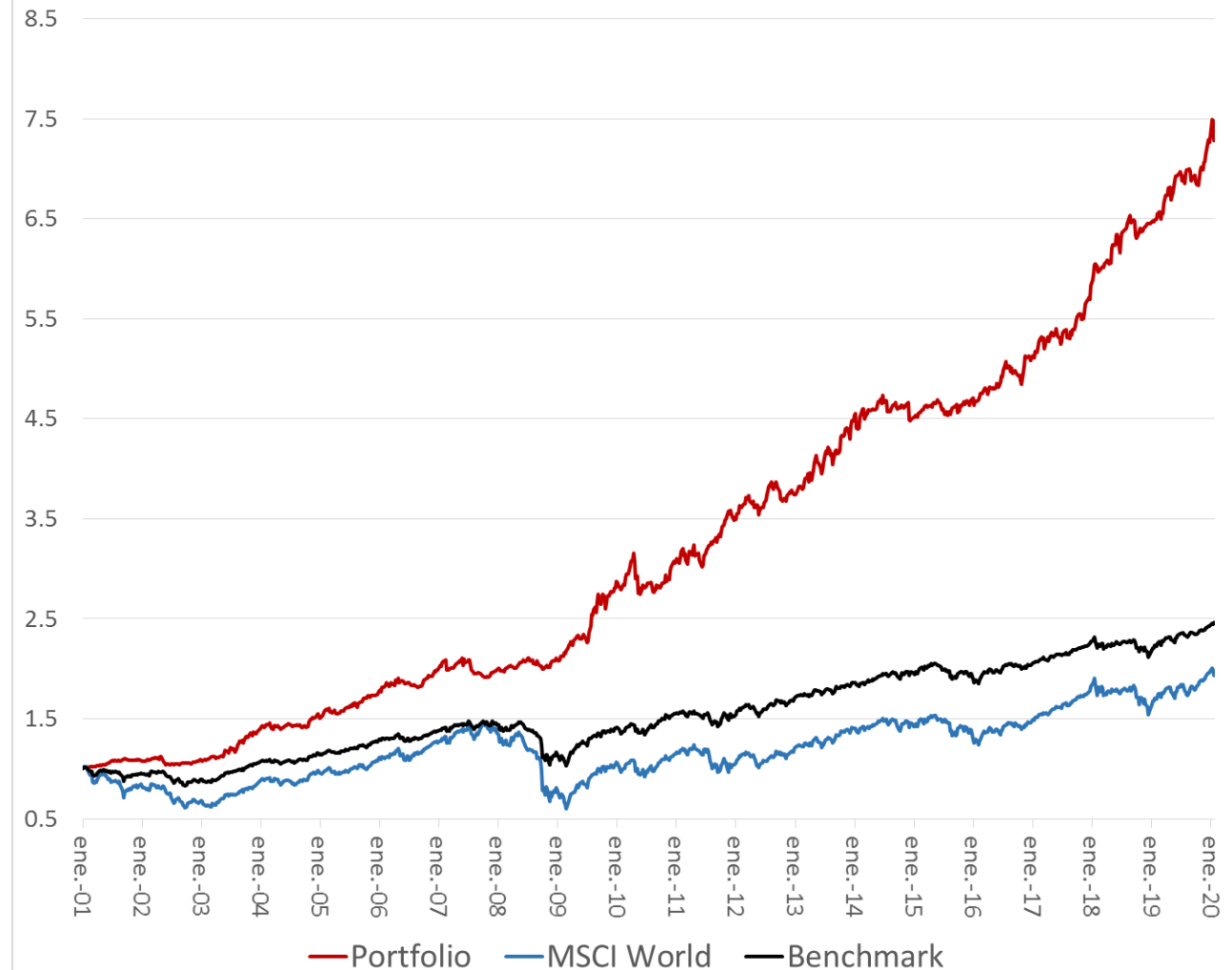
Realized Active Risk: Knowing when to "Turn off" the Strategy



Back tested Results: Compensating our baseline model

Backtested Results of Momentum Model

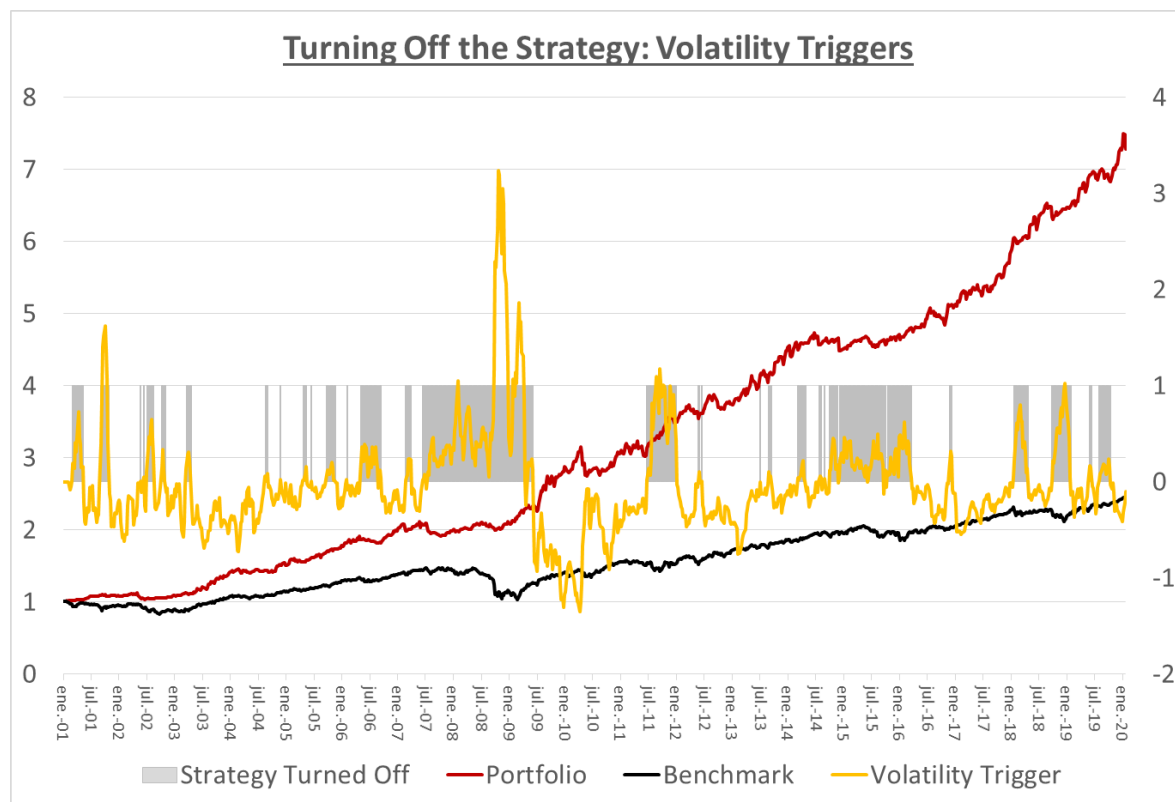
Benchmark: Portfolio of 50/50 Stocks and Bonds



The Key point of the Momentum Model: Knowing when to “Turn off” the Strategy

Volatility Indicator

Momentum Strategies are known for having sudden moments of underperformance. We have constructed a **Volatility Indicator** which tries to capture pick ups in Volatility that “Turn off” the strategy → **Avoid Losing Cumulative Alpha by being invested in the wrong moment**



The Results of the Momentum Model in 2018

Since 2018

Risk Metrics

	Cumulative Returns	TAA Benchmark	MSCI World
Annualized Return	12.69	3.71	4.11
SD of Yearly Return	6.17	8.66	13.23
Risk Adjusted Return	2.06	0.43	0.31
Max DD	-2.93	-8.55	-19.18
Calmar Ratio	4.32	0.43	0.21
Max Weekly Loss	-2.70	-3.15	-5.75
Max Weekly Gain	2.48	1.96	4.33
Annualized Alpha	8.21		
Active Risk	8.88		
Information Ratio	92.49%		

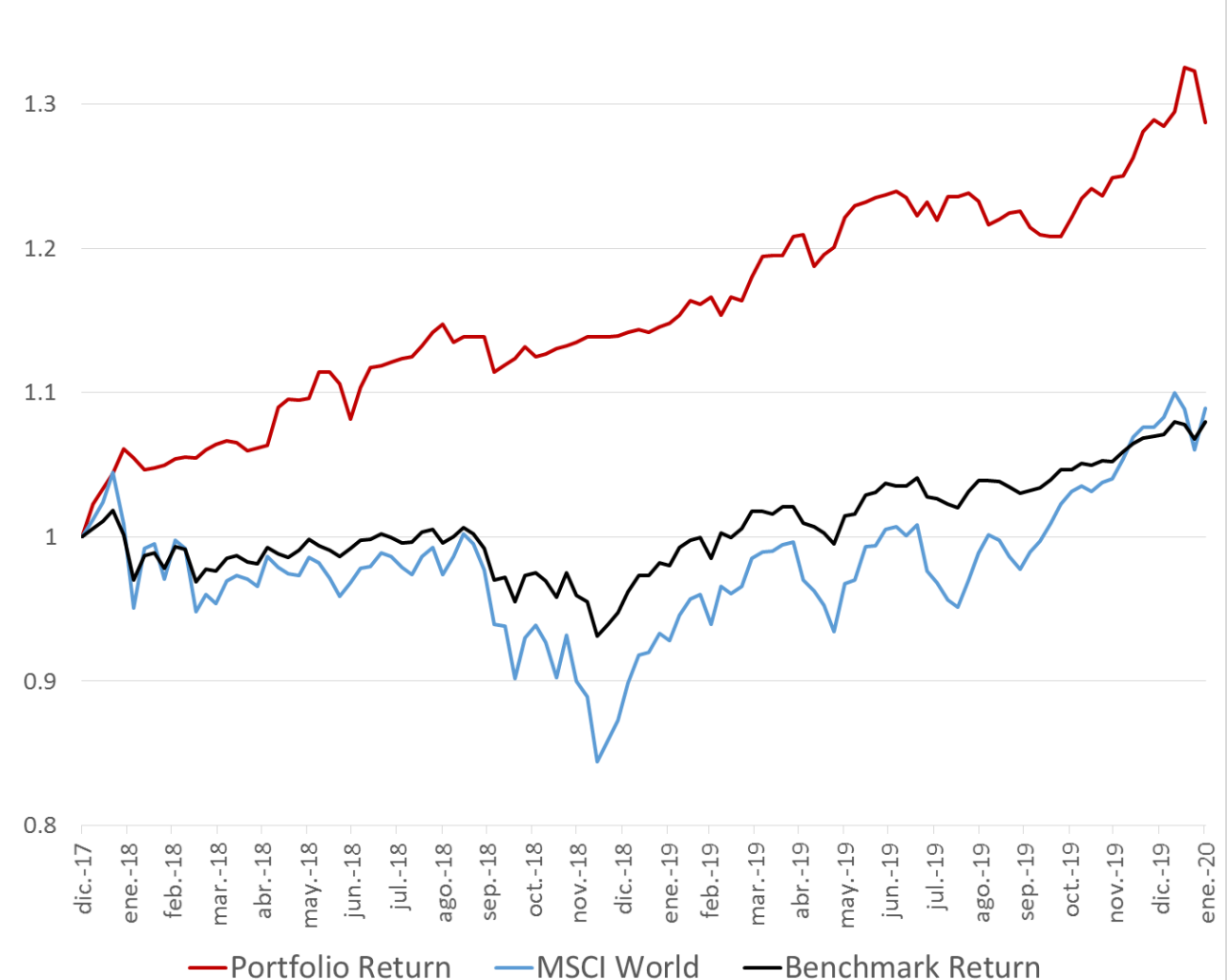
Realized Active Risk: Knowing when to "Turn off" the Strategy



Back tested Results: Compensating our baseline model

Backtested Results of Momentum Model

Benchmark: Portfolio of 50/50 Stocks and Bonds



Combining the Strategies: Economic Regime Model and Momentum Model

What is the idea behind integrating both Models?

Economic Regime Based Model

We infer what **Economic Regime** we are in and set strategies that should perform well across:

- Asset Classes
- Factor Tilts
- Curve Positioning
- Geographic Regions

Limitations:

- Cannot control for Policy Action
- Does not capture market shocks unless its reflected in data

Contrarian Model (Composite Sentiment Indicator)

In moments of Financial Stress we take a **Contrarian** view to Neutralize the portfolio and avoid noise.

- We try to avoid rapid recovery in Risk Assets
- Do not assume Active Risk in periods of high volatility

Limitations:

- It's a contrarian Indicator so therefore it is rarely activated

Momentum Strategy adjusted for "Quality"

Conversely we assume we try to capture the Market Regime which assets are dictating by favoring those assets that are outperforming through **Momentum**

The focus is to compensate performance in periods of Market Noise

Limitations:

- Momentum Strategies are known for having periods of large Drawdowns

Combining both the Strategies

By combining both Models the idea is to have a consistent framework where the deficiencies of both models try to offset each other

Limitations:

- Correlation between both strategies is time varying
- Subject to the limitations of both models

Modulating each Strategy Based on Correlations

By changing the weight that we assign to each model across time based on the **correlation** of both strategies we are able to:

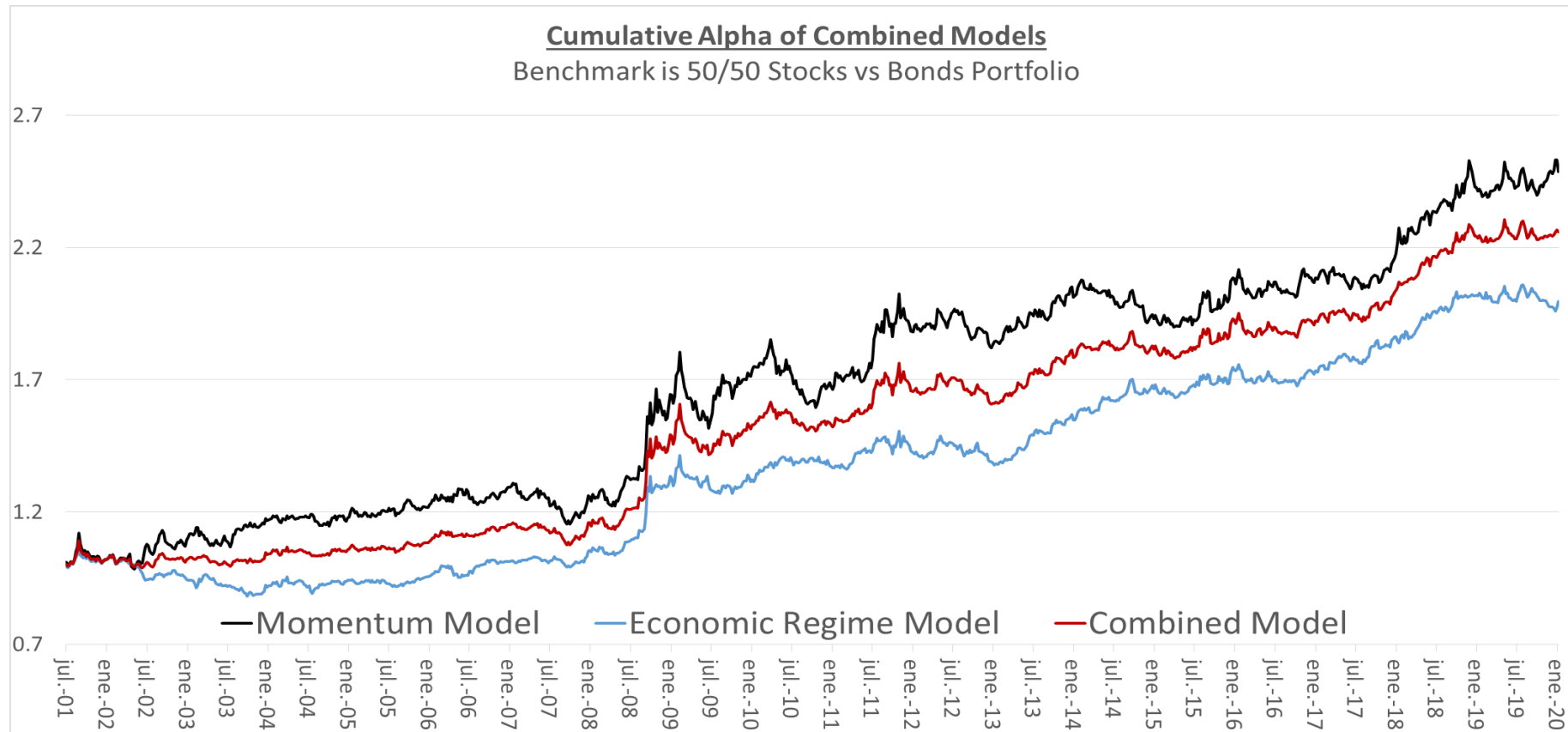
- Assign less weight to one of the strategies when correlations are high as to not duplicate the models
- Reward the Momentum Strategy when Correlations are low to capture Market Turn

Combining Both Models: Economic Regime Model + Momentum Model

What are the results?

- The Combination of Both Strategies provide superior Information Ratio vs a 50/50 Benchmark of Stocks and Bonds
- The Correlation between both strategies also works when intended, in inflection points.
- The portfolio adds risk more swiftly that the Economic Regime Model

Finally Modulating
for Dynamic
Weights



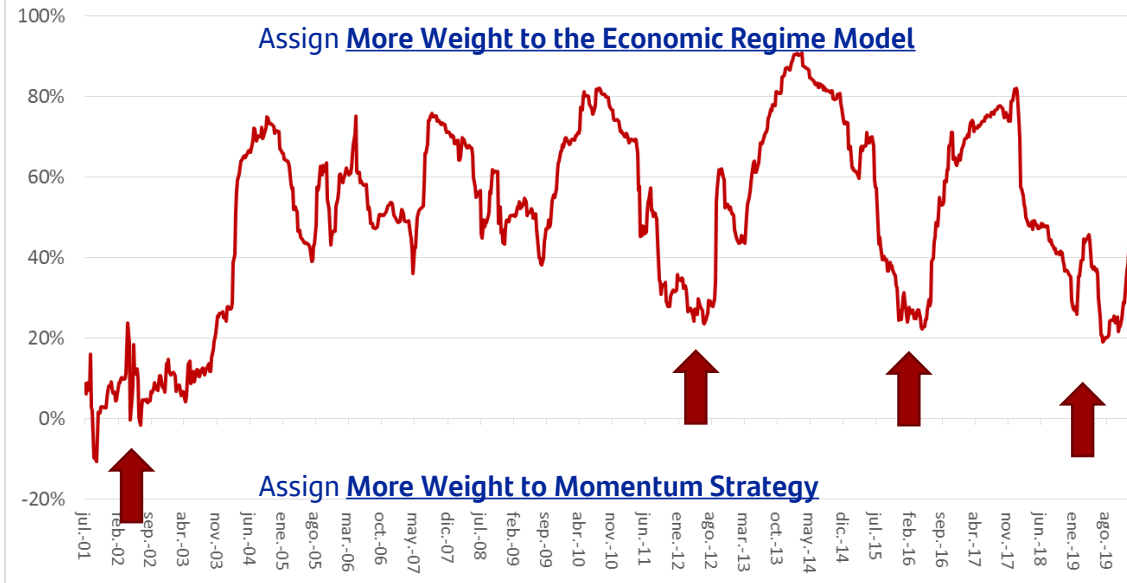
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Dynamic Combined Model Results

Rolling Correlation of Combined Models:

Economic Regime Model and Momentum Model



Risk Metrics

	Economic Model	Momentum Model	Dynamic Combined Model	TAA Benchmark	MSCI World
Annualized Return	9.60	10.92	10.74	4.83	3.51
SD of Yearly Return	5.45	7.90	7.08	8.66	16.68
Risk Adjusted Return	1.76	1.38	1.52	0.56	0.21
Max DD	-8.20	-12.86	-9.91	-30.29	-58.76
Calmar Ratio	1.17	0.85	1.08	0.16	0.06
Max Weekly Loss	-5.49	-5.87	-4.91	-10.79	-20.07
Max Weekly Gain	3.60	4.95	4.95	6.99	12.37
Annualized Alpha	4.87	5.44	4.91		
Active Risk	8.15	8.77	6.46		
Information Ratio	59.74%	62.00%	75.91%		

Back tested Results: Compensating our baseline model

Cumulative Alpha of Combined Models with Dynamic Weights

The Weights Assigned to Each Model Depends on the Correlation

